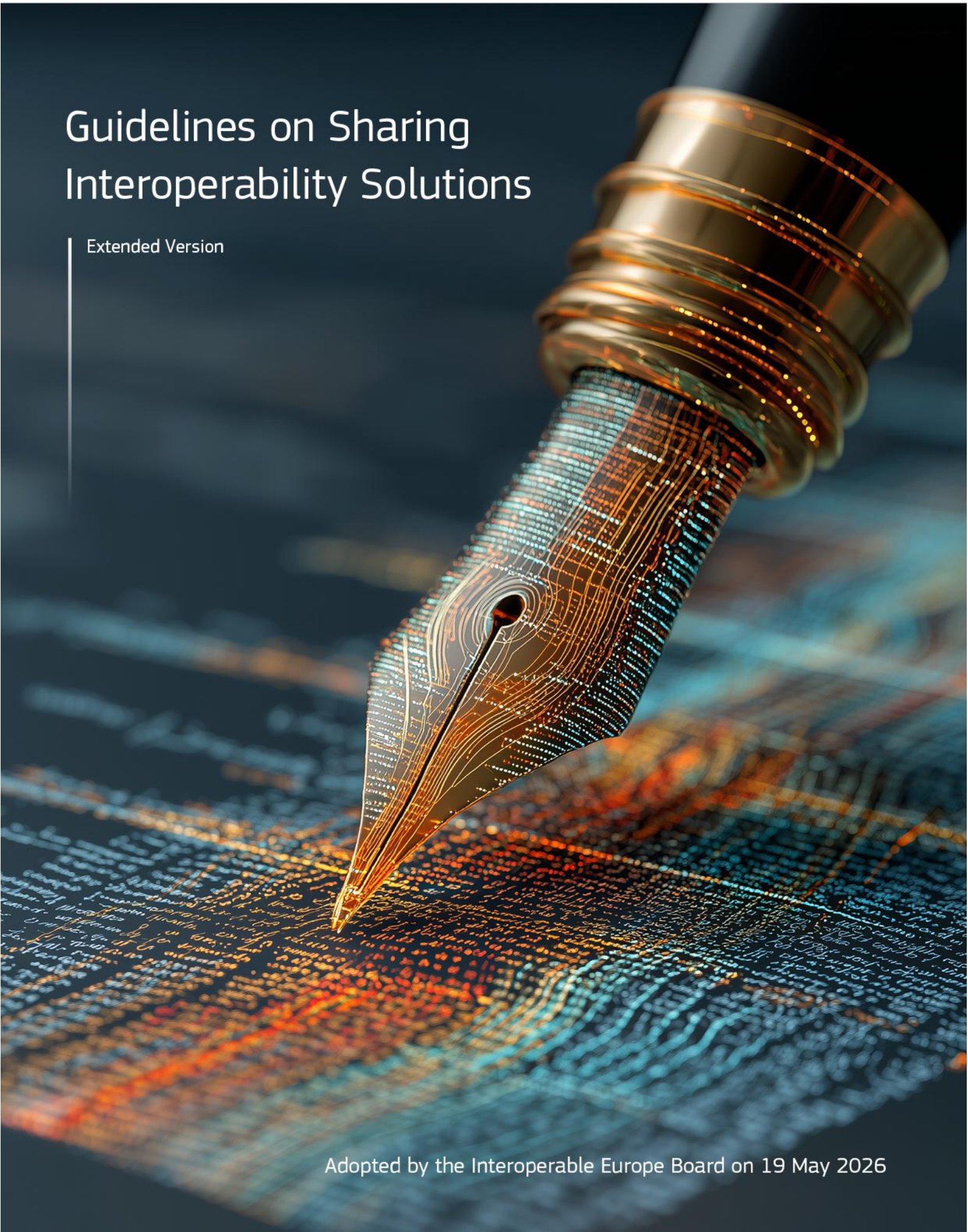
European Commission

interoperable europe

Guidelines on Sharing Interoperability Solutions

Extended Version



Adopted by the Interoperable Europe Board on 19 May 2026

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EXECUTIVE SUMMARY

The Interoperable Europe Act (IEA, the Act) establishes a legal and organisational framework to strengthen cooperation between public administrations across the European Union in the delivery of interoperable digital public services. A central element of this framework is the sharing and reuse of interoperability solutions, enabling public organisations to build once, and reuse across borders, sectors, and levels of government. Sharing interoperability solutions across borders can also contribute to broader digital sovereignty objectives, by helping public organisations avoid unnecessary dependency, retain control over their digital choices, and build on reusable solutions developed and governed within the European public sector.

In accordance with **Article 4(7)** of the IEA, these **Guidelines on the Sharing of Interoperability Solutions** provide practical support to support Union institutions and public sector bodies in implementing the sharing and reuse provisions of the Act. Their purpose is to help Union entities and public sector bodies apply those provisions in a consistent, proportionate and operationally workable manner across different administrative, legal and technical contexts.

The Guidelines are available in two complementary versions: an extended version, which provides the full guidance framework, and a short version, which presents the essential operational pathways for applying Articles 4 and 8 in practice. Together, they are intended to support both quick operational use and more detailed reference.

The Guidelines are legally grounded, but they **do not create additional obligations** beyond those laid down in the IEA and other applicable law. Rather, they distinguish between the legal requirements arising under the IEA, route-specific operational conditions, recommended implementation practices to support effective application, and optional good practices that entities may use to strengthen reuse, collaboration and long-term sustainability.

The core of the document is a **pathway-based structure** that translates the IEA into practical operational routes. These pathways are organised in an operational order rather than in the sequence of the legal provisions, and they are not sequential or mutually exclusive. More than one pathway may apply over the lifecycle of the same interoperability solution. The six pathways covered in these Guidelines are: prioritising solutions without restrictive licensing terms where equivalent; publishing interoperability solutions through the Interoperable Europe Portal or connected portals, catalogues and repositories; direct sharing upon request; sharing adapted interoperability solutions; cost-sharing for future developments; and making portals, catalogues and repositories interoperable with the Interoperable Europe Portal.

The document is supported by governance guidance, lifecycle framing, procurement-related guidance and annexes addressing issues such as maintenance responsibilities, metadata and publication context, and the documentation of adapted solutions. To complement the pathway-based guidance, the Guidelines include **illustrative practices and examples drawn from experiences across Member States and Union entities**. These examples are intended to support understanding and practical application.

The Guidelines are intended to be a **reference document**, supporting public organisations across diverse policy and service domains. They should be read in conjunction with existing and forthcoming guidance developed in the context of the Interoperable Europe framework made available through the Interoperable Europe Portal.

ABOUT THE GUIDELINES

The Interoperable Europe Act (IEA, the Act)[1] establishes a reinforced cooperation framework to support the development and delivery of interoperable cross-border public services within the Union. In particular, Article 4 makes the sharing and reuse of interoperability solutions the general rule for Union entities and public sector bodies where such solutions support trans-European digital public services and are requested by another such entity or body, subject to the conditions and exceptions laid down in the Act. These Guidelines are intended to support public organisations[2] in applying that framework in practice.

Interoperability solutions – such as legal frameworks, reference architectures, technical specifications, standards, software and associated documentation – help public organisations address legal, organisational, semantic, and technical interoperability requirements in a coherent manner. Articles 4 and 8 of the Act provide the basis for their sharing, publishing, reuse and visibility, including through the Interoperable Europe Portal[3] and connected portals, catalogues, and repositories. Article 4(7) mandates the development of guidelines to support the implementation of these provisions.

These Guidelines are presented in two complementary versions: this extended version, which provides the full guidance framework, and a short version, which focuses on the operational pathways for applying Articles 4 and 8. The short version offers a streamlined entry point for day-to-day implementation, while this extended version provides the broader legal, organisational and operational context, including governance, procurement-related guidance, examples from practice, and supporting annexes.

Beyond their primary purpose, the Guidelines may also serve as reference material for public organisations seeking to develop their own guidelines, templates or other supporting materials for sharing and reuse, adapted to their legal frameworks, governance arrangements, administrative context, technical environment, security requirements, resource constraints and operational needs. More broadly, they may serve as a voluntary reference beyond the specific scope of Article 4 – including for solutions that do not support trans-European digital public services – and can guide public organisations in organising the sharing, reuse, adaptation and governance of interoperability solutions across different settings, including within Member States and across national, regional, and local public administrations. They do not, however, aim to address business models or organisational design in detail.

Purpose and objectives of the Guidelines

The Guidelines aim to:

- translate the legal requirements and mechanisms of Articles 4 and 8 of the IEA into clear and operational implementation guidance;
- support public organisations in sharing and reusing interoperability solutions in a consistent, transparent, and legally compliant manner;

[1] Regulation - EU - 2024/903. In this document, 'Interoperable Europe Act' is abbreviated as 'IEA'. References to 'the Act' should be understood as references to the Interoperable Europe Act, unless otherwise indicated.

[2] For the purposes of these Guidelines, 'public organisation' is used as a collective shorthand to refer, as appropriate in context, to Union entities and public sector bodies within the scope of the Interoperable Europe Act. 'Sharing entity' and 'reusing entity' describe the role performed by such actors in a given pathway. These expressions are used for readability and functional description only, and do not replace, extend or modify the terminology or legal categories used in the Act.

[3] Interoperable Europe Portal

- promote the reuse of existing solutions across administrative levels and borders, thereby reducing duplication of effort and improving the use of public resources;
- provide practical guidance for integrating interoperability solutions into public procurement and tendering processes.

In doing so, the Guidelines seek to foster a common understanding and shared practices among public organisations, while allowing sufficient flexibility to account for different administrative, legal, and organisational contexts.

Target audience

These Guidelines are primarily addressed to:

- Union institutions, bodies, and agencies,
- national, regional, and local public administrations of the Member States,
- public officials and practitioners involved in digital transformation, interoperability, information and communication technologies, and public procurement,
- stakeholders responsible for the development, maintenance, sharing, adaptation or reuse of interoperability solutions.

They may also be of interest to private sector actors, open source communities and other parties involved in the development or implementation of interoperability solutions for the public sector, insofar as their activities relate to the sharing and reuse frameworks described herein.

Structure of the Guidelines

The Guidelines are structured as follows:

- Chapter 1 – Governance Model and Roles: clarifies the roles and responsibilities established under the IEA and describes optional internal roles that public organisations may consider supporting implementation.
- Chapter 2 – Operational Entry Points and Lifecycle of Sharing and Reuse: presents a high-level lifecycle view of sharing and reuse, from identifying an interoperability solution to sharing, reuse, maintenance, and evolution.
- Chapter 3 – Practical Pathways for Sharing Interoperability Solutions: provides the core operational guidance in these Guidelines. It explains how Articles 4 and 8 of the IEA may be applied in practice through six pathways.
- Chapter 4 – Integrating Interoperability Solutions into Public Procurement: explains how sharing and reuse considerations can be reflected in procurement processes, in line with the IEA and the applicable Union public procurement framework.
- Annexes: provide supporting material, including additional technical context, references and operational guidance on matters such as maintenance responsibilities, metadata and publication context, voluntary good practices, and the documentation of adapted interoperability solutions.

These Guidelines are available in both an extended version and a short version. The short version is available here: The Guidelines | Interoperable Europe Portal. It focuses on the practical pathways set out in Chapter 3 and is intended as a streamlined companion to this extended version.

The Guidelines do not replace existing legal obligations under Union or national law, nor do they introduce new binding requirements beyond those established by the Interoperable Europe Act. They should be read and applied in conjunction with applicable legal frameworks, security requirements, data protection rules, and sector-specific regulations.

KEY CONCEPTS AND DEFINITIONS

Interoperability

Interoperability refers to the ability of Union entities and public sector bodies to interact with each other by sharing data, information and knowledge through digital processes, in line with the applicable legal, organisational, semantic and technical requirements. Cross-border interoperability refers to such interactions when they take place across Member States' borders.

Trans-European digital public service

A trans-European digital public service is a public service provided by Union entities or public sector bodies to one another, or to natural or legal persons in the Union, that requires interaction across Member State borders through network and information systems. The sharing obligation under Article 4 applies only to interoperability solutions supporting such services.

Interoperability solutions

An interoperability solution is a reusable asset that supports cross-border interoperability. Under the Act, the term may cover legal, organisational, semantic, or technical requirements. In practice, it may include a conceptual framework, guideline, reference architecture, technical specification, standard, service, application, or documented technical component such as source code.

Interoperability solutions may vary significantly in terms of maturity. For the purposes of these Guidelines, public organisations may distinguish between three indicative levels:

- Experimental solutions: early-stage assets, prototypes or pilots, with limited documentation and evolving functionality;
- Operational solutions: solutions in active use within one or more public organisations, with sufficient stability and partial documentation;
- Reusable solutions: mature solutions designed for reuse, with stable functionality, structured documentation, and clear governance and maintenance arrangements.

Interoperable Europe Solutions

Interoperable Europe solutions[4] are interoperability solutions identified at Union level, recommended by the Interoperable Europe Board, and published by the Commission on the Interoperable Europe Portal, in accordance with the Act. They form part of the coordinated Union-level interoperability ecosystem.

Sharing

Sharing means making an interoperability solution available to a Union entity or public sector body under Article 4. This may take place through direct sharing upon request under Article 4(1), or through publication on the Interoperable Europe Portal or on a connected portal, catalogue, or repository under Article 4(3). Where Article 4(1) applies, sharing includes the interoperability solution together with the technical documentation and, where applicable, version history, documented source code, and references to open standards or technical specifications used.

Reuse

Reuse means that an entity other than the original developer or sharing entity takes up an interoperability solution for its own purposes. Reuse may consist in implementing the solution as it is, or in adapting it to legal, organisational, semantic, or technical needs, in accordance with the reuse conditions.

[4] Interoperable Europe Solutions | Interoperable Europe Portal

Adaptation

Adaptation means modifying an interoperability solution to meet specific needs. Under Article 4(4), a Union entity, public sector body, or third party reusing an interoperability solution may adapt it, unless third-party intellectual property rights restrict adaptation. Where the original solution was made public under Article 4(3), the adapted solution must be made public in the same way.

Reuse conditions

Where an interoperability solution is shared directly under Article 4(1), the sharing entity must specify any conditions applying to reuse under Article 4(2). In these Guidelines, these are referred to as reuse conditions. They may include any guarantees provided regarding cooperation, support, or maintenance. They may also include exclusion of the sharing entity's liability in case of misuse by the reusing entity. Article 4(2) requires the sharing entity to specify any guarantees provided, but does not require ongoing support or maintenance by default.

Interoperable Europe Portal

The Interoperable Europe Portal is the platform provided by the Commission as a single point of entry for information related to the cross-border interoperability of trans-European digital public services. It provides access both to Interoperable Europe solutions and to other interoperability solutions, including those made available under Article 4(3) and those published on connected portals, catalogues, or repositories.

EU Open Source Solutions Catalogue

The EU OSS Catalogue[5] is a catalogue available through the Interoperable Europe Portal. It is a federated platform and repository for open source solutions developed by or for, or useful for, public sector bodies across the European Union. It aggregates OSS from national, regional and local catalogues using public APIs and may also include certain standalone or noteworthy third-party open source solutions where the published conditions are met.

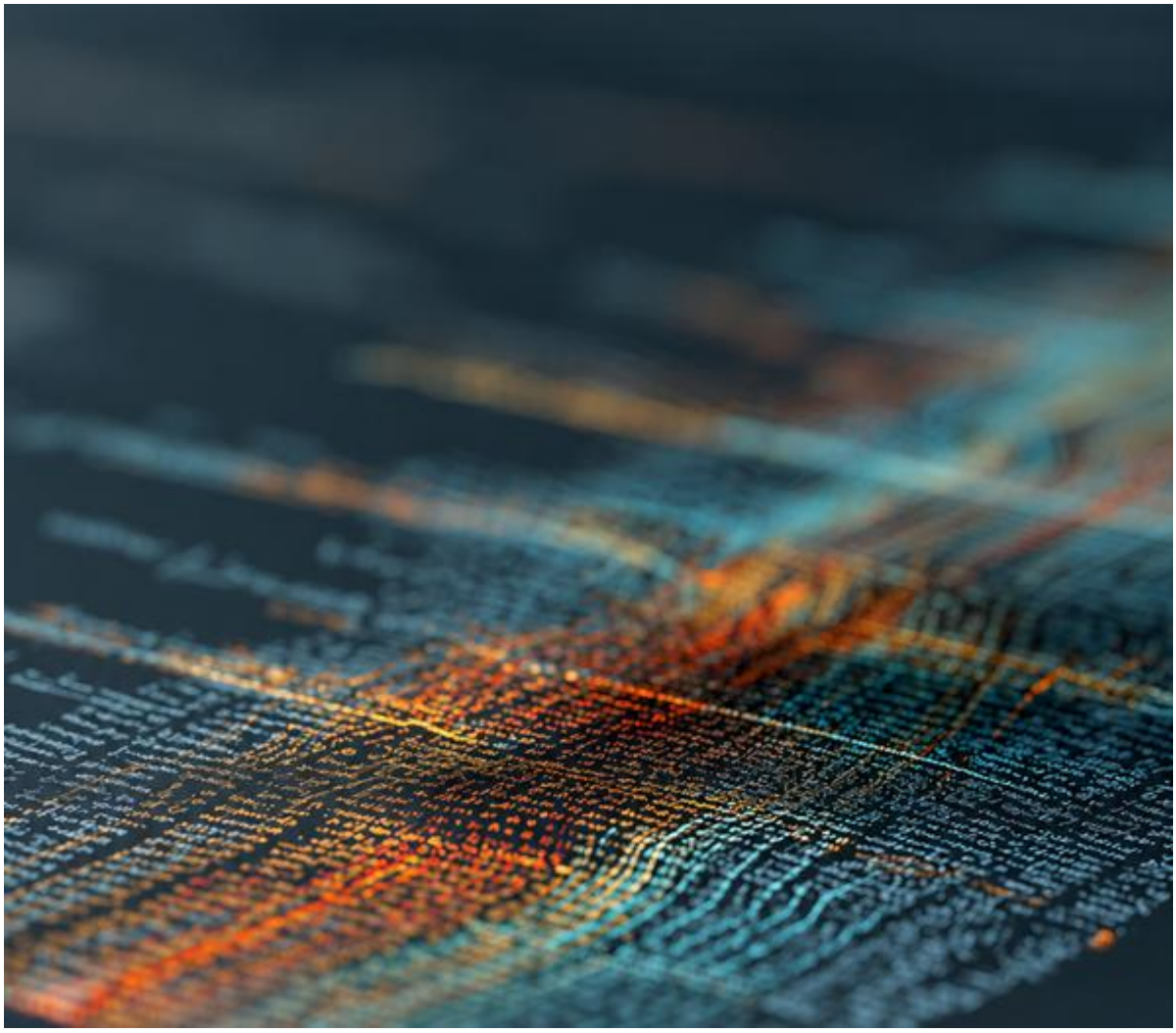
Solutions accessible through the Interoperable Europe Portal

Under Article 8(3), an interoperability solution may be made accessible through the Interoperable Europe Portal only if certain conditions are met. Such solutions must not be subject to third-party rights preventing distribution and use, must not contain personal data or confidential information, must have a high level of alignment with Interoperable Europe solutions, must use a licence allowing reuse at least by other Union entities or public sector bodies or be issued as open source, and must be regularly maintained under the responsibility of the owner.

Connected portal, catalogue, or repository

A connected portal, catalogue, or repository is a portal, catalogue, or repository that is linked to the Interoperable Europe Portal through an operational arrangement that makes relevant interoperability solutions accessible through that Portal in a stable and maintainable manner, in line with Article 8(4). It may support, depending on its function, discoverability, descriptive metadata, publication, access, collaboration, or the maintenance of authoritative artefacts and versions. It may be operated by a Union entity or public sector body, jointly with others, or by a third party on its behalf. Where such a portal, catalogue, or repository collects open source solutions, it must allow for the use of the European Union Public Licence (EUPL)[6].

[5] EU Open Source Solutions Catalogue | Interoperable Europe Portal
[6] EUPL | Interoperable Europe Portal



CHAPTER 1. GOVERNANCE MODEL AND ROLES

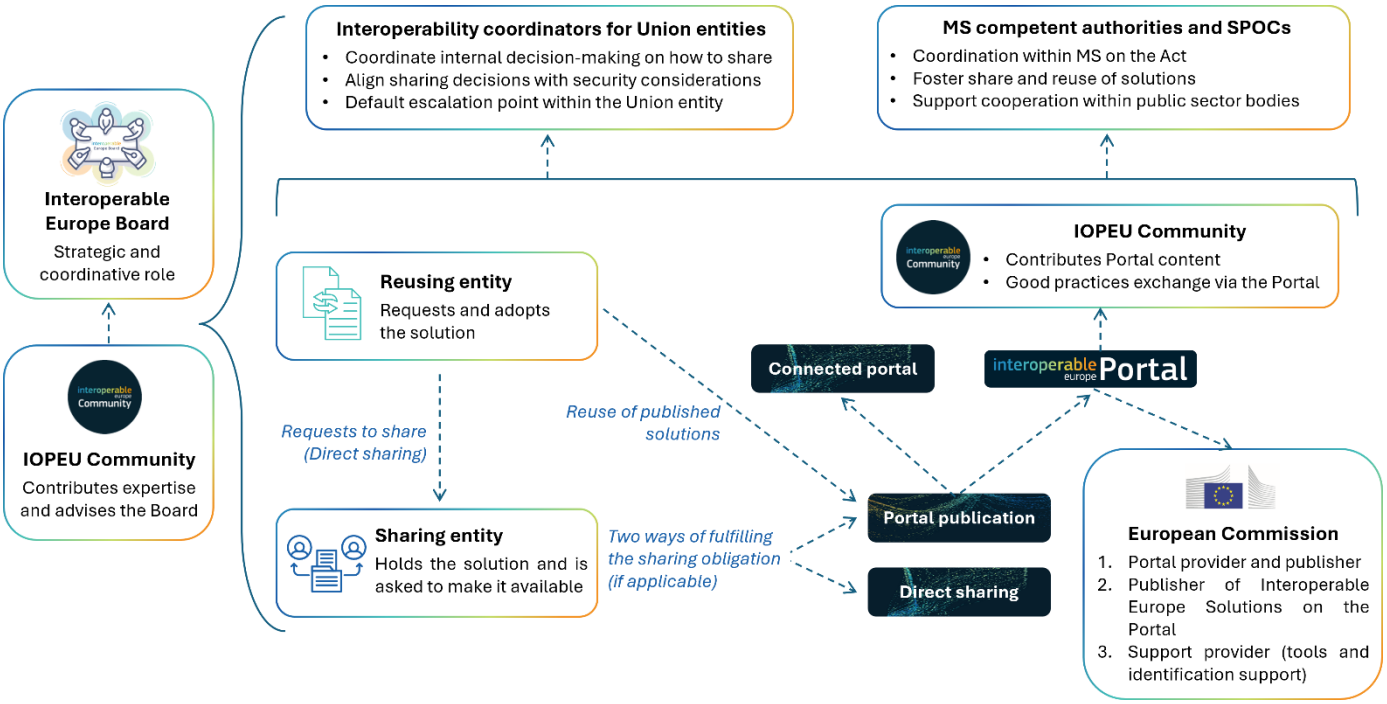
This chapter sets out a practical governance model for sharing and reusing interoperability solutions between Union entities and public sector bodies. It clarifies roles established under the Act and describes optional internal roles that public organisations may consider to support implementation across the sharing and reuse process.

1.1. Governance model at a glance

For each interoperability solution, a Union entity or public sector body may consider ensuring that the following elements are in place:

- **an owner in the sharing entity** (accountable for decisions, sharing, publication, and maintenance);
- **a defined intake channel** for requests and reuse questions (for example, a functional mailbox, service desk, or repository channel depending on the nature of the solution);
- **named points of contact** for legal/licensing, security, and technical stewardship if applicable;
- **a documented decision trail** for key decisions (share, publish instead of direct sharing, refuse with reasons, set reuse conditions).

This is consistent with the Act's approach: obligations sit with the Union entity or public sector body as such, while operational delivery relies on internal roles and processes.



1.2. Roles and responsibilities under the Interoperable Europe Act

1.2.1. Sharing entity

The **sharing entity** is the Union entity or public sector body that holds the interoperability solution and is asked to make it available.

Legal duties (where Article 4(1) applies)

The sharing entity must make available to any other Union entity or public sector body that requests it an interoperability solution supporting a trans-European digital public service, including:

- technical documentation,

and where applicable,

- version history,

- documented source code for software,
- references to open standards or technical specifications used.

Grounds for not sharing (Article 4(1) exceptions)

The obligation to share does not apply where the interoperability solution:

- falls outside the scope of the sharing entity's public task, as defined by law or binding rules, or by established administrative practice, provided that the scope is transparent and subject to review;
- is constrained by third-party intellectual property rights that restrict sharing for reuse;
- is excluded or restricted on grounds of sensitive critical infrastructure protection, defence interests or public security, including national critical infrastructure.

Reuse conditions and autonomy assessment (Article 4(2))

Where the public organisation shares directly under Article 4(1), it must specify any conditions that apply to reuse, including any guarantees on cooperation, support and maintenance. Those conditions may include exclusion of liability in case of misuse by the reusing entity.

Before adoption, the reusing entity must, upon request by the sharing entity, provide an assessment of its ability to manage autonomously:

- cybersecurity,
- the evolution of the reused solution.

Option to fulfil the sharing obligation by publishing (Article 4(3))

The sharing entity may fulfil the obligation to make an interoperability solution available by publishing the relevant content on:

- the Interoperable Europe Portal,
- a portal, catalogue, or repository connected to it.

At the sharing entity's request, the Commission must publish the relevant content on the Interoperable Europe Portal.

Where the obligation is fulfilled via publication, the sharing entity is not required to specify reuse conditions under Article 4(2).

Cost-sharing and open source prioritisation (high-level responsibilities)

- The sharing and reusing entities may conclude an agreement on sharing the costs for future developments, financially or in-kind contributions. (Article 4(5))
- When deciding on implementation, Union entities and public sector bodies must prioritise non-restrictive licensing terms (such as open source) where solutions are equivalent against objective criteria (functionalities, total cost, user-centricity, cybersecurity, or other relevant criteria). (Article 4(6))

1.2.2. Reusing entity

The **reusing entity** is the Union entity or public sector body requesting and adopting the interoperability solution.

Core responsibilities

- Formulate a request and identify the intended use (which trans-European digital public service is supported).
- Implement the solution in line with the sharing entity's reuse conditions (where applicable).

Autonomy assessment (Article 4(2))

If the sharing entity requests it, the reusing entity must provide an assessment covering its ability to manage autonomously:

- cybersecurity,
- the evolution of the reused interoperability solution.

Adaptation and publication of adapted solutions (Article 4(4))

A reusing entity (and, where relevant, a third party reusing the solution) may adapt the solution in line with the applicable licensing conditions, unless restricted by third-party intellectual property rights. Where the original interoperability solution is made public under Article 4(3), the adapted solution must be made public in the same way. Applicable licensing conditions may affect the practical conditions of adaptation and publication, but they do not displace that statutory rule.

1.2.3. Interoperable Europe Board

The Board's role is strategic and coordinative, rather than operational delivery, and it does not extend to handling individual sharing cases.

Key responsibilities relevant to these Guidelines

- The Board shall adopt guidelines on sharing interoperability solutions.
- The Board proposes measures to foster sharing and reuse.
- The Board may propose that the Commission publish other interoperability solutions on the Portal or have them referred to on the Portal.
- The Board may recommend interoperability solutions as Interoperable Europe solutions and may withdraw such recommendations in accordance with agreed criteria.

1.2.4. Member State competent authorities and single points of contact

Each Member State designates one or more competent authorities and a single point of contact (Article 17).

In the context of sharing and reuse, the single point of contact's tasks include, among others:

- coordination within the Member State on questions related to the Act;
- fostering share and reuse of interoperability solutions through the Interoperable Europe Portal or another relevant portal;
- contributing country-specific knowledge to the Interoperable Europe Portal;
- supporting cooperation with public sector bodies in other Member States.

Having a single point of contact has proven effective in streamlining internal governance and simplifying the handling and documentation of sharing requests.

Practical application in these Guidelines

Public sector bodies may consider using the single point of contact as an enabling function, for example to:

- identify suitable solutions already shared;
- connect reusing entities to owners/maintainers;
- support cross-border coordination where multiple authorities are involved.

1.2.5. Interoperability coordinators for Union entities

Union entities that regulate, provide, or manage trans-European digital public services must designate an **interoperability coordinator** under oversight of their highest level of management.

In practice, interoperability coordinators are well-placed to:

- coordinate internal decision-making on whether to share directly or publish;
- align sharing decisions with internal security and legal functions;
- act as the default escalation point within the Union entity for complex reuse cases.

This does not create new tasks for interoperability coordinators beyond the Act but describes a workable operational allocation.

1.2.6. European Commission

In the context of the Guidelines, the Commission has three distinct operational roles.

(a) Portal provider and publisher

The Commission must provide the Interoperable Europe Portal as a single point of entry and ensure it has the functions set out in Article 8(1) (including access to solutions and knowledge exchange features).

When a sharing entity requests it, the Commission must publish the relevant content on the Interoperable Europe Portal for the purpose of Article 4(3).

(b) Publisher of Interoperable Europe solutions

The Commission publishes Interoperable Europe solutions on the Portal, in open and machine-readable formats (with accessibility requirements).

(c) Support provider (tools and identification support)

The Commission provides technical tools to support interoperability assessments, including an online tool to facilitate completion and publication.

The Commission also provides support in identifying non-restrictively licensed solutions, linked to support measures.

In their role as secretariat of the Interoperable Europe Board, the Commission supports Member States in coordinating their public investments in interoperability solutions through the Interoperable Europe Agenda process.

1.2.7. Interoperable Europe Community

The Interoperable Europe Community may contribute practical experience and good practices relevant to the sharing and reuse of interoperability solutions, in particular through consultations and Portal-related activities.

1.3. Optional internal roles to support implementation

The Act assigns obligations to entities. To deliver them, public organisations may consider assigning some of the profiles listed below, taking into account their own structure, size, and internal arrangements. For each interoperability solution intended for sharing or publication, public organisations may in particular consider the following roles.

1.3.1. Proposed roles

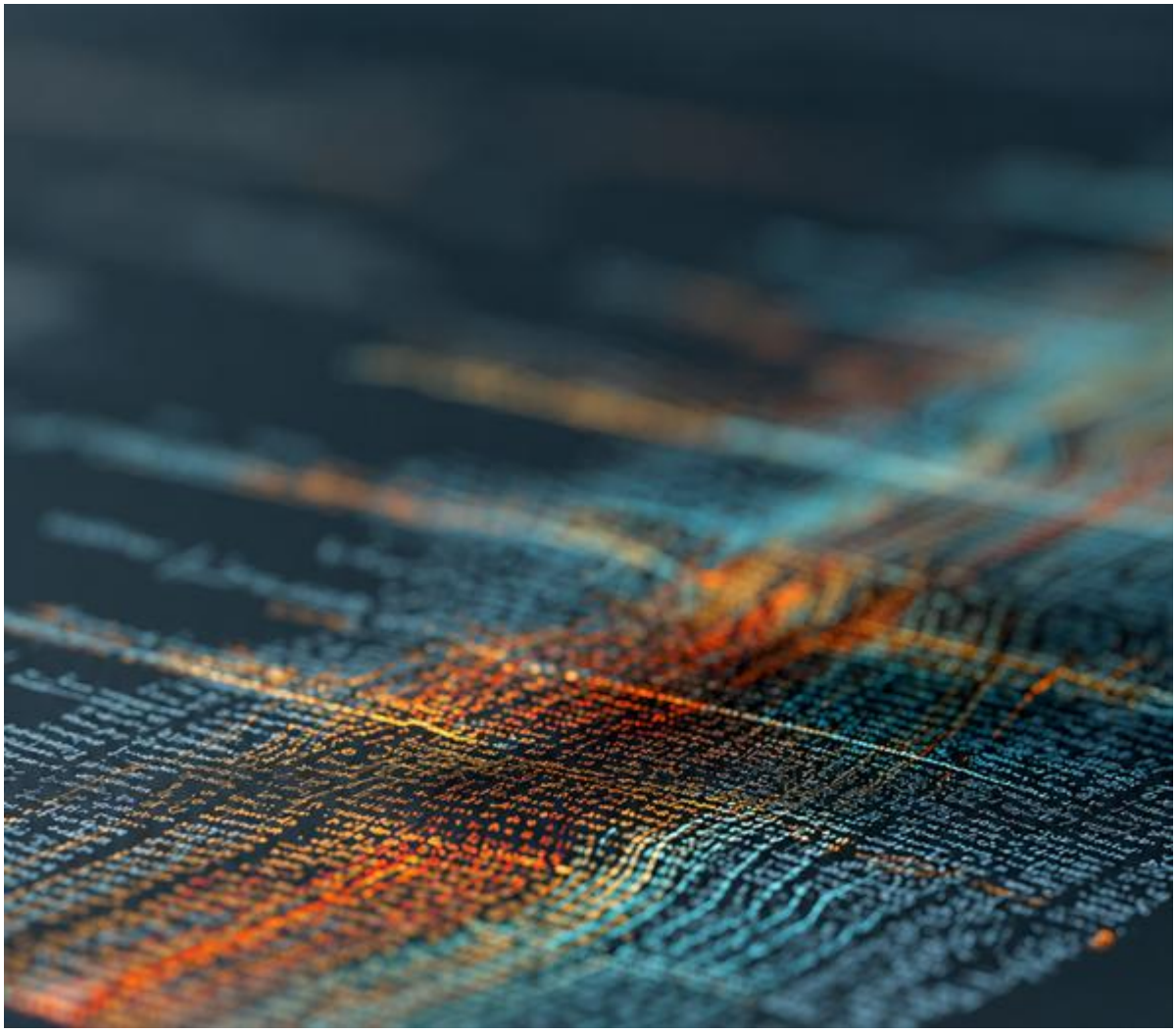
- **Solution owner**: decides on sharing or publication within the applicable framework, sets any reuse conditions, oversees the sharing process, and ensures maintenance of the solution, including where it is accessible through the Interoperable Europe Portal.
- **Technical steward**: prepares the shareable package; manages versioning and release information; and provides technical clarification during reuse in line with the declared support stance.

- **Portal publishing contact**: manages metadata for publication mechanics (uploading or linking).

1.3.2. Control roles that may be involved as needed

These functions do not need to be full-time roles, especially for smaller public organisations, but may be made available through internal teams or shared services.

- **Legal and licensing contact**: assesses third-party intellectual property constraints relevant to Article 4(1)(b) and adaptation under Article 4(4); and confirms that the selected licence permits reuse in line with reuse conditions and Portal conformance criteria.
- **Security contact**: advises on whether access to a solution should be excluded or restricted on grounds of public security, defence or critical infrastructure protection; and reviews cybersecurity implications of reuse, including, where requested, the reusing entity's autonomy assessment.
- **Data protection and confidentiality contact**: ensures that content made publicly available through the relevant publishing portal, including the Interoperable Europe Portal or any portal, catalogue or repository connected to it, does not include personal data or confidential information, in accordance with applicable data protection, confidentiality and publication rules..
- **Procurement or finance contact**: supports the preparation of cost-sharing arrangements for future developments under Article 4(5) and ensures compatibility with internal budgeting and procurement rules. This contact may also provide advice on prioritising the implementation of reusable interoperability solutions within the public procurement process.

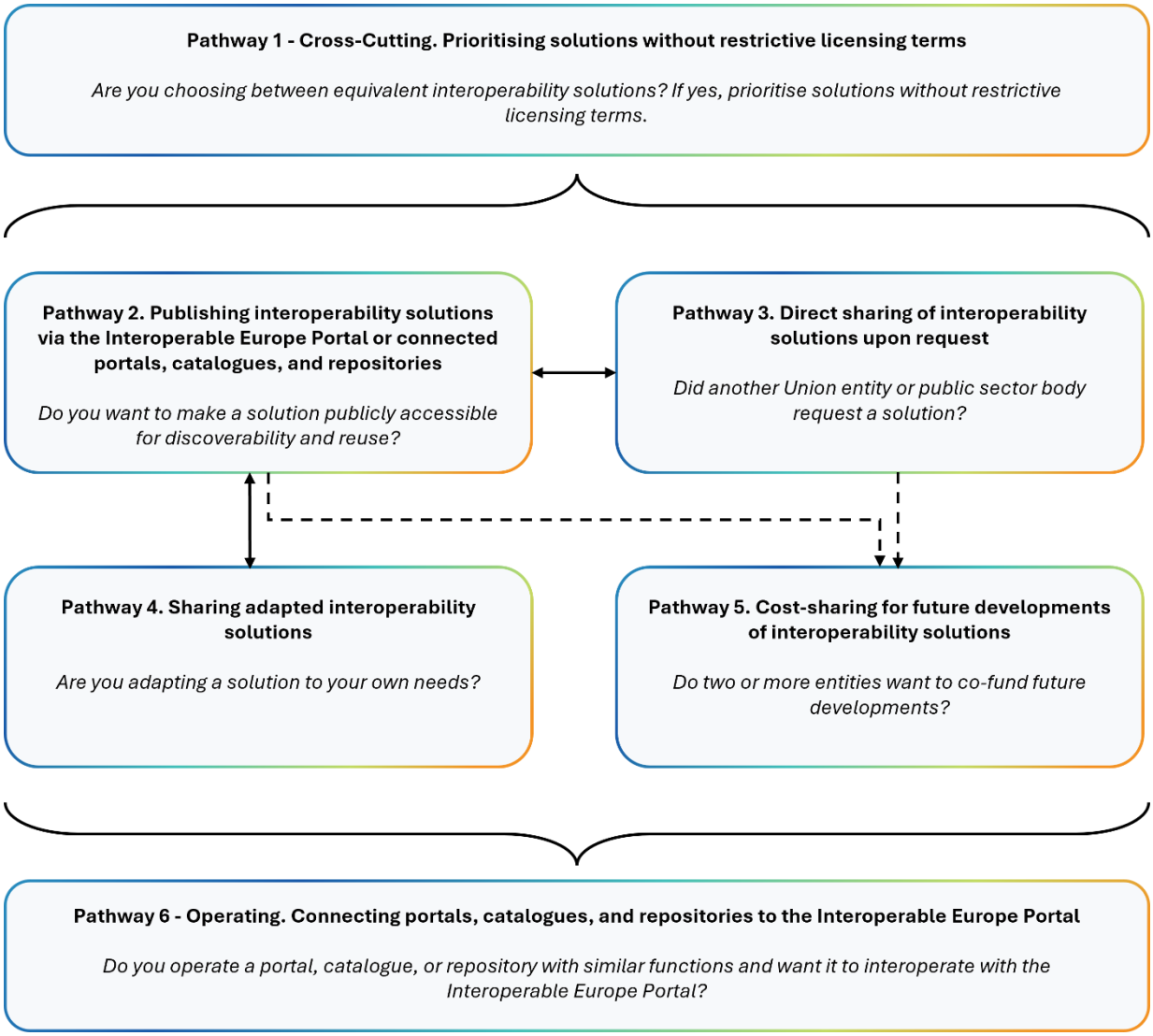


CHAPTER 2. OPERATIONAL ENTRY POINTS AND LIFECYCLE FOR SHARING AND REUSE

This chapter helps public organisations identify their starting point and navigate the operational lifecycle of sharing and reuse under the Interoperable Europe Act. It is organised around practical entry points, rather than the sequence of the legal provisions.

2.1. Identifying the relevant pathway

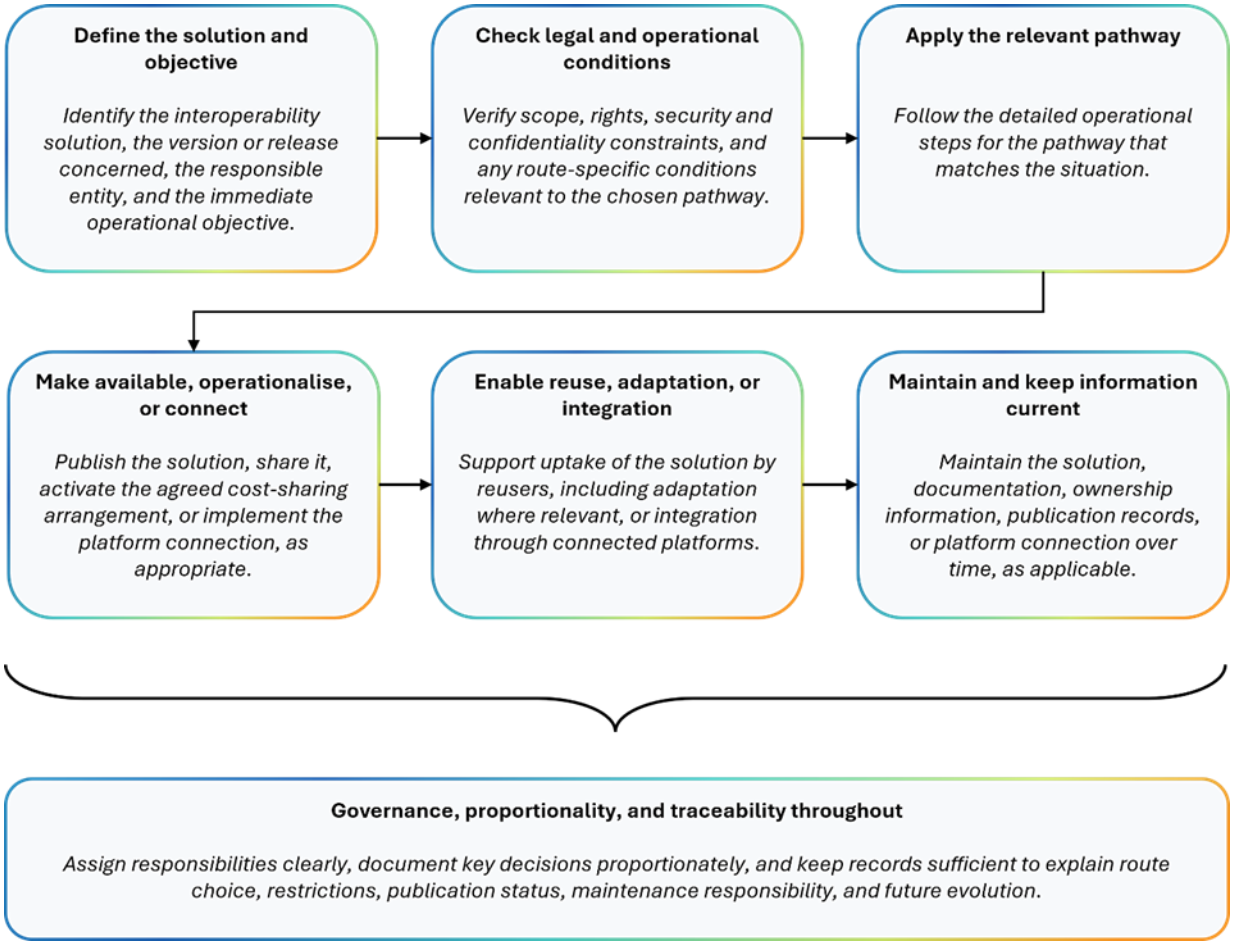
Union entities and public sector bodies may need to fulfil obligations under the IEA in different situations: when deciding how to implement a solution, when making a solution publicly accessible, when responding to a request from another public organisation, when adapting and sharing a reused solution, when organising future developments jointly, or when operating a portal, catalogue or repository with similar functions. The first task is therefore to identify the entry point that best matches the immediate legal and operational situation.



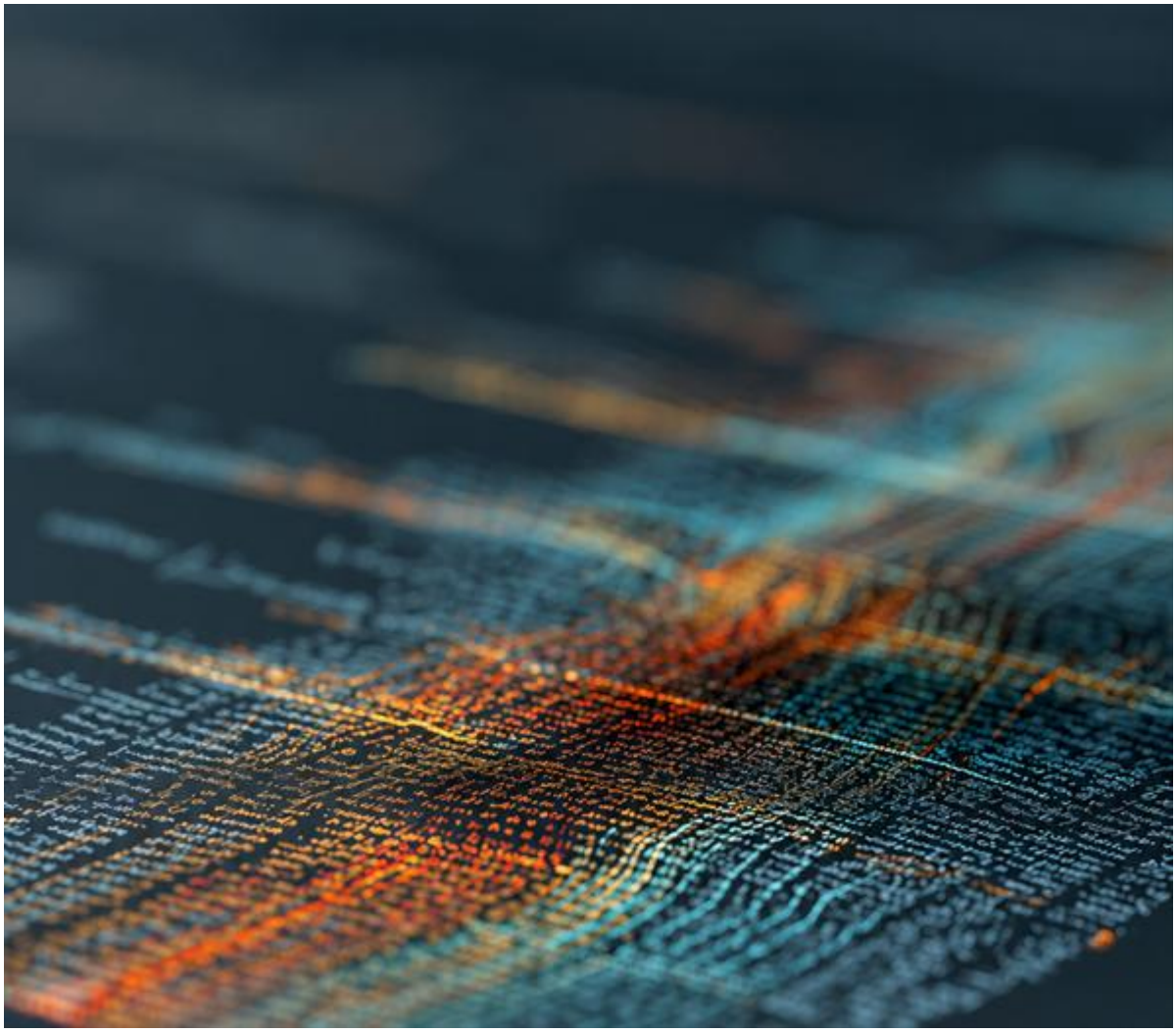
The figure above is a navigation aid and it does not create a hierarchy between pathways. Rather, it helps readers identify which pathway in Chapter 3 is most relevant to their starting point. Once the relevant pathway has been identified, the common lifecycle stages can be used to structure implementation in a coherent and proportionate manner. Depending on the type of interoperability solution and the maturity of the reuse context, some steps will be central, while others will be only marginally relevant. The pathways should therefore be applied proportionately. Their application should also take account, where relevant, of the broader ecosystem in which the solution is developed, maintained, governed, reused or supported.

2.2. Common lifecycle stages after pathway identification

Once the relevant pathway has been identified, public organisations can use a common set of lifecycle stages to structure implementation. These stages apply flexibly and proportionately across the different pathways. They are intended to support planning, coordination and traceability, not to impose a single mandatory process.



Stages may overlap, repeat, or lead back to an earlier step as interoperability solutions are reused, adapted, and evolved.



CHAPTER 3. PRACTICAL PATHWAYS FOR SHARING INTEROPERABILITY SOLUTIONS

This chapter presents six practical pathways for applying Articles 4 and 8 of the Interoperable Europe Act to the sharing and reuse of interoperability solutions.

The pathways are neither sequential nor mutually exclusive, and more than one may apply over the lifecycle of the same interoperability solution.

Navigating the Pathways

To support navigation, each pathway follows the same structure:

- **When the pathway applies**: Each pathway begins with a short description explaining the situation in which it is relevant. This helps public organisations identify which pathway best corresponds to their legal and operational context.
- **Legal basis overview**: The applicable provisions of the IEA are presented upfront. This overview sets out the legal foundation for the pathway before moving into practical guidance.
- **Legal basis translated into steps**: The legal provisions are then translated into a sequence of implementation steps, highlighted using a green banner as shown below:

> **Article number**
>
> Explanation.

- **Implementation guidance**: For each step, practical guidance is provided to support implementation. This guidance is not mandatory and is intended to help public organisations apply the legal provisions in a proportionate and operationally realistic manner. Throughout, the wording 'must', 'shall', and 'should' (and similar variations where relevant) are reserved for legal requirements; whereas 'may' and 'could' (and similar variations where relevant) for operational good practices.
- **Step outputs**: Each step concludes with a clearly identified output, highlighted using a blue banner as shown below. These outputs summarise the tangible result of the step and help understand what should be achieved before moving on.

Output of this step: Explanation.

- **'In focus' boxes**: Where relevant, the pathways include description of real-life practices or examples from Member States or Union entities. These examples are illustrative and aim to show how similar situations have been handled in practice. They are not exhaustive and are not included for every step.

In focus

Example.

3.1. Prioritising solutions without restrictive licensing terms (Article 4(6))

This pathway applies when a public organisation:

- is deciding whether to implement an interoperability solution;
- is comparing candidate interoperability solutions and one or more options may be available without restrictive licensing terms;
- is determining the licensing approach for an interoperability solution that it intends to make available for reuse, where that choice forms part of a decision on the implementation of the solution.

In the context of the Act, solutions without restrictive licensing terms will in practice most often be open source solutions. Their licences generally permit use, reuse, modification and redistribution, subject, where applicable, to conditions such as attribution or copyleft. This provides the legal clarity and predictability needed for reuse, adaptation, integration and redistribution across different administrative, legal and technical contexts. Examples include software components supporting digital public services, such as tools for digital forms, API management, identity integration, document exchange, workflow management, metadata catalogues, interoperability registries and semantic mappings made available under open source licences. This legal clarity facilitates reuse across different administrative, legal and technical contexts, can reduce implementation effort and duplication, and facilitate collaborative improvement and wider dissemination. For that reason, Article 4(6) requires Union entities and public sector bodies to prioritise such solutions where they are equivalent in terms of the objective criteria laid down in the Act.

Solutions without restrictive licensing terms are not limited to software. Depending on the asset concerned, they may also include documentation, semantic assets, reference architectures or other reusable artefacts made available under open content or open data licences that permit reuse. Conversely, an interoperability solution, may still fall within the scope of the Act even if it carries restrictive licensing terms, since the definition of interoperability solution is functional and does not depend on the licensing model. This may include, for example, standards or specifications developed by third parties or standardisation bodies.

However, restrictive terms may limit reuse, adaptation, and integration, for example by creating legal uncertainty, requiring individual negotiations, restricting modification or redistribution, or creating dependency on a specific provider. This may reduce practical interoperability and slow cross-border adoption. Where restrictive terms or service dependence cannot be avoided, the assessment should explicitly consider exit, dependency management and, where relevant, data portability.

Interoperability solutions can be shared and reused through different delivery models, including on-premise deployment and ‘as-a-service’ provision. In such cases, reuse, adaptation and integration are typically governed by the applicable usage licence terms and/or service contracts, which define what is permitted (e.g. configuration, extension, integration), and under which conditions (e.g. access, support, service levels, security, and continuity).

Legal basis

- **Prioritisation rule**: When deciding on the implementation of interoperability solutions, Union entities and public sector bodies must prioritise **solutions that do not carry restrictive licensing terms, such as open source solutions**, where solutions are equivalent in terms of functionalities, total cost, user-centricity, cybersecurity, or other relevant objective criteria (Article 4(6)).
- **Commission support**: The Commission shall provide support in identifying such interoperability solutions, as provided for in Article 9 (Article 4(6)).

3.1.1. Implementation guidance

3.1.1.1. Step 1. Identify the relevant implementation decision under Article 4(6)

Article 4(6)

The obligation applies when deciding on the **implementation** of interoperability solutions.

Before applying the prioritisation rule, the first step may be to identify the specific decision that gives rise to the application of Article 4(6).

Article 4(6) applies to decisions concerning the implementation of interoperability solutions. Such decisions do not necessarily concern the replacement or selection of an entire system. In practice, particularly in the context of large or mandated systems, interoperability choices are often made at architectural, governance or design level within an existing system.

Accordingly, the relevant decision may include:

- **selecting an interoperability solution to implement**;
- deciding whether to **develop a new interoperability solution** or to **replace or adapt** an existing one;
- choosing, within an existing or mandated system, a data model, interface standard, semantic asset or reference architecture that performs an **interoperability function**;
- determining, as part of such a decision, the **licensing terms** under which the interoperability solution will be made available.

Clarifying the decision at the outset helps ensure that the assessment of equivalence and the application of the prioritisation rule are integrated into ordinary IT governance, procurement, and software development processes, rather than treated as an isolated or ad-hoc exercise. In practice, this also means considering from the outset whether an equivalent solution already exists under a non-restrictive or open source licence, or whether a new solution could itself be developed and made available as open source.

Possible actions may include:

- defining the functional, technical, and organisational requirements that the solution must meet;
- identifying the main constraints relevant to the context, including budget, timeline, integration/infrastructure environment, accessibility, security, data protection, and maintenance needs;
- integrating this decision point into existing governance, procurement or software development lifecycle processes, so that prioritisation under Article 4(6) can be applied systematically where relevant.

Output of this step: a clearly formulated implementation decision, accompanied by requirements list against which candidate interoperability solutions can be assessed.

In focus

The **OSOR Handbook: Open Source Software in Public Administration** is a practical guidance document developed for the European Commission's Open Source Observatory (OSOR) to **support public administrations in working with open source software**.

The handbook provides concise, experience-based guidance for administrations seeking to adopt or strengthen open source practices based on extensive experience and community consultation. It addresses the full lifecycle of open source use in the public sector, including identifying and evaluating solutions, development and procurement, participation in open source communities, funding, licensing, and organisational structures.

3.1.1.2. Step 2. Identify candidate solutions, starting with reusable solutions already available, and establish a shortlist for further assessment

Article 4(6)

The **Commission shall support** identifying such solutions as provided for in Article 9.

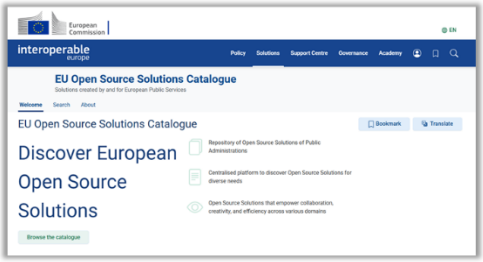
When identifying candidate solutions, the public organisation may take a broad and pragmatic approach. Candidate solutions may include:

- **existing solutions within its own organisation**, including solutions already in use or under development;
- **interoperability solutions made available by other public sector bodies**, including those accessible through the Interoperable Europe Portal or connected portals, catalogues or repositories;
- **other publicly available solutions** on catalogues or repositories developed by or for the public sector that may meet the stated requirements.

The Commission supports this identification process through the Interoperable Europe Portal and related discovery tools, including the EU OSS Catalogue. Further support may also be provided through the measures and services referred to in Article 9. This support is intended to facilitate discovery and comparison, without prescribing a specific source or tool.

The EU OSS Catalogue aggregates open source solutions through a federated model and covers open source software from national, regional and local catalogues as well as certain standalone or noteworthy third-party open source solutions that meet the published conditions. Such catalogues are supporting tools and are not legally mandated sources.

The EU Open Source Solutions Catalogue (EU OSS Catalogue)



The EU OSS Catalogue **federates** national and local catalogues that meet specific technical and governance criteria.

Open source solutions can be displayed and filtered through the advanced search available.



Where a potentially suitable solution is known to exist but is not publicly accessible, the public organisation may consider seeking it through the direct sharing arrangements described in these Guidelines in Section 3.3. Where a solution is or will be made publicly accessible for reuse, the publication guidance in these Guidelines may be relevant (Section 3.2).

This consideration supports efficiency, interoperability and reuse, while remaining subject to applicable legal, technical, and organisational constraints.

At this stage, the entity may also exclude options that are manifestly unsuitable and establish a shortlist for fuller assessment under the next step. This does not yet amount to a final selection decision.

Output of this step: a shortlist of candidate solutions, including any reusable solutions already identified.

In focus

Support or endorsement by a recognised entity, such as an **Open Source Programme Office** (OSPO) or a national or EU institutional body, can enhance the credibility of an open source solution and facilitate cross-border sharing and adoption. Code managed within such frameworks is typically developed under **high standards of quality, transparency, usability, privacy and societal value**.

For instance, the European Commission OSPO supports the adoption and governance of open source solutions within the European Commission. Public sector bodies are encouraged to engage with established OSPOs.

3.1.1.3. Step 3. Assess equivalence and operational feasibility against objective criteria

Article 4(6)

Prioritisation applies where **solutions are equivalent** in terms of functionalities, total cost, user-centricity, cybersecurity, or other relevant objective criteria.

Equivalence does **not** require that solutions be identical. It requires a level of functional and operational comparability that is sufficient to meet the stated needs of the public organisation.

Before taking a final selection decision, the public organisation should assess both equivalence and the operational feasibility of shortlisted candidate solutions in a manner that is objective, transparent, and proportionate. In practice, this assessment can be carried out using a simple comparison matrix covering, as appropriate:

- **functionality**: whether the solution meets the defined requirements, and any identified gaps;
- **total cost**: including development or acquisition costs, operational costs, maintenance, and exit or transition costs;
- **user-centricity**: usability, accessibility, language support, and alignment with user needs;
- **cybersecurity**: patching processes, dependency management, vulnerability handling, and security governance;

- **other relevant objective criteria**, such as sustainability, standards compliance, portability, interoperability maturity, architectural fit, and privacy and data-protection implications, including data minimisation and proportionality where relevant.

Under the 'other relevant objective criteria' referred to in Article 4(6), public organisations may, where relevant, assess the extent to which a candidate solution supports long-term operational autonomy, reduces the risk of lock-in, enables knowledge retention within the organisation, and facilitates transition to alternative maintainers or providers. These considerations may be applied as criteria for assessing equivalence and long-term viability, not as additional legal prerequisites.

Where relevant to the context, the assessment may also consider factors commonly associated with open source solutions and other solutions without restrictive licensing terms, such as:

- the potential for **reuse and wider deployment** across public organisations;
- the **availability and quality** of technical and non-technical documentation;
- the clarity of **governance** and **maintenance** arrangements;
- the handling of **updates, bug fixes, and security issues**;
- where maintenance depends materially on a **community**, indicators of that community's activity, responsiveness, continuity, and long-term sustainability;
- **source code** availability, where relevant.

Where relevant, public organisations may support this assessment through a **preliminary market analysis or feasibility study**, in line with applicable rules and practices. Depending on the solution and community or supplier ecosystem, risks can include higher onboarding and integration effort, limited or unclear corrective and evolutive maintenance arrangements, insufficient documentation, limited technical support or service options, language barriers, and limited evidence of reliability, performance, scalability or security in comparable deployments. These factors are relevant for determining whether solutions are truly equivalent in practice, and for selecting appropriate mitigation measures (e.g. support contracts, managed services, assurance testing, documentation improvement, or community engagement).

Where one or more shortlisted solutions are available without restrictive licensing terms, the public organisation may also confirm, before final selection, that **it can implement, operate, maintain, and secure the solution in a sustainable manner**. This may be particularly relevant where a shortlisted solution is open source, will be adapted locally, or depends in whole or in part on external maintainers or communities. For non-software solutions, equivalent questions of stewardship, update handling, documentation, access and governance should be considered proportionately. Possible actions may include:

- assigning an **internal owner** responsible for the solution,
- defining **maintenance and support processes**, including how updates and fixes will be handled,
- establishing a **security** monitoring, reporting, and patching workflow,
- confirming that the public organisation has the **technical environment and internal capacity** needed to host, integrate, and operate the solution,
- ensuring that the **documentation** needed to understand, configure, maintain, and, where relevant, adapt the solution is available,
- clarifying, where relevant, how **external contributions** are handled and how improvements will be contributed back to an upstream project or community.

Operational feasibility may, where appropriate, also be supported through contractual arrangements with external providers responsible for installation, operation or maintenance of the solution. Where such arrangements exist, the assessment could also consider dependency management, documentation quality, knowledge transfer and exit readiness. Architectural

choices may, where feasible, support reuse and adaptation, for example through modular design that allows individual components to be shared or reused independently.

Output of this step: a documented equivalence assessment and operational adoption plan, saved as part of the decision record.

In focus

When public organisations assess standards or technical specifications as part of this equivalence assessment, **structured methodologies** such as **Common Assessment Method for Standards and Specifications** (**CAMSS**) may be used to support transparent and comparable evaluation that supports interoperability and helps preventing vendor lock-in. Where a complete CAMSS assessment is already available, it can be reused as documented justification when preparing a solution for sharing. If no prior assessment exists, public organisations may conduct their own evaluation using the EIF Scenario or request assistance from the CAMSS team.

Selecting standards already assessed through CAMSS not only supports compliance but also facilitates maintenance, adaptation and evolution of solutions over time. Including the **CAMSS assessment in the sharing package** provides concrete evidence that the solution is based on objectively evaluated standards, offering assurance to reusing public organisations. For long-term sustainability, CAMSS assessments should be revised whenever standards or APIs are updated.

3.1.1.4. Step 4. Apply the prioritisation rule, take the selection decision, and record the rationale

Article 4(6)

If equivalent, the non-restrictively licensed solution, such as open source, shall be **prioritised**.

Prioritising the use and sharing of interoperability solutions that do not carry restrictive licensing terms may also support digital sovereignty, by facilitating inspection, reuse, adaptation and, where applicable, collaborative maintenance across public organisations.

Once the prioritisation rule is applied, the public organisation may record the rationale for its decision in a concise and structured way. In practice, this could include:

- a brief explanation of **why the solutions were assessed as equivalent or not equivalent**;
- a description of **how licensing restrictiveness was assessed** for the solutions considered;
- where relevant, a short explanation of why the selected solution was considered **operationally suitable**.

This record supports transparency, consistency and later review, without requiring extensive justification.

Output of this step: a decision record documenting the application (or non-application) of the prioritisation rule under Article 4(6).

3.1.1.5. Step 5. Choose a licence and manage compatibility

Except where Article 8(3)(d) or Article 8(4) is relevant, the Act does not prescribe a specific licence to choose.

Where a public organisation develops or adapts an interoperability solution and intends to make it available for reuse, it may consider whether the solution can be made available without restrictive licensing terms and could document the licensing approach chosen. If broad reuse across public organisations is intended, this may include, in particular:

- assess whether third-party rights, dependencies, or contractual arrangements affect the **licensing options** available;
- select a licence or set of terms that fits the **legal, technical, and organisational context** and supports **lawful reuse**;
- document any **licence compatibility considerations** and any relevant notices or obligations;
- clarify, where relevant, how **external contributions** are accepted and how contributions to upstream projects will be handled.

Clear and transparent licensing arrangements help reduce legal uncertainty and facilitate reuse by other public sector bodies.

The **European Union Public Licence (EUPL)**, developed by the European Commission as the first EU-created open source licence, is widely used within the EU public sector and is designed to support sharing and reuse across public organisations operating under different national legal frameworks. In particular, the EUPL is recognised for:

- its legal consistency with the copyright law of all EU Member States, which reduces legal uncertainty for cross-border reuse;
- its availability in all official EU languages, facilitating use by public organisations without the need for additional legal translation;
- its explicit licence-compatibility provisions, which enable interoperability with several other commonly used open source licences and support the combination and reuse of software components across projects.

The EUPL may be a suitable option for many public sector solutions, especially in cross-border contexts, because it is available in all official EU languages and includes licence-compatibility provisions. Other licences remain possible where the legal or technical context justifies them. Third-party licence compatibility may need to be assessed under applicable intellectual property law.

Where a Union entity or public sector body operates a portal, catalogue or repository with similar functions to the Interoperable Europe portal and that platform collects open source solutions, it must allow for the use of the EUPL. This requirement concerns the operation of the platform and does not establish the EUPL as a default or mandatory licence for all open source software.

Output of this step: a documented licensing approach, including the licence or licensing terms selected and any key compatibility considerations.

In focus

To support the licence assessment, the Licensing Assistant, including its licence compatibility checker, is available on the Interoperable Europe Portal.

3.1.2. Monitoring and performance indicators

The IEA does not require the monitoring of specific indicators for the prioritisation of non-restrictively licensed solutions under Article 4(6). Public sector bodies may nonetheless monitor progress to assess the effectiveness of reuse and open source adoption.

Basic indicators:

- number or percentage of implementation decisions in which an equivalence assessment considered at least one solution without restrictive licensing terms;
- number or percentage of equivalent cases in which a solution without restrictive licensing terms was prioritised.

Advanced indicators:

- estimated time or cost savings achieved through reuse of existing solutions rather than new development;
- number of interoperability solutions made available by the public organisation under non-restrictive licensing terms;
- number of reuses or adaptations of such solutions by other public organisations, where known;
- compliance rates with licensing, security, and maintenance requirements.

Monitoring should support learning and improvement, not create additional compliance burdens.

3.1.3. Risks and mitigation actions

Applying Article 4(6) may give rise to legal, technical, organisational, and operational risks if equivalence assessments are weak or if solutions are selected without sufficient implementation planning. Typical risks include superficial comparison exercises, over-reliance on licensing labels without adequate assessment of functionality, total cost, user-centricity or cybersecurity, unclear third-party rights, inadequate maintenance arrangements, or insufficient internal capacity to operate and secure the selected solution.

These risks can be mitigated by defining objective criteria in advance, documenting the equivalence assessment, involving legal, security, procurement, and technical functions where appropriate, checking third-party rights and dependency compatibility early, and confirming maintenance and governance arrangements before implementation. Where the selected solution is open source, proportionate vulnerability monitoring, patch management, and security review should form part of the operational baseline.

3.2. Publishing interoperability solutions via the Interoperable Europe Portal or connected portals, catalogues, and repositories (Article 4(3))

This pathway applies when:

- the sharing entity decides to fulfil Article 4(1) by publishing content,
- the sharing entity proactively publishes content to enable reuse at scale,
- an adapted interoperability solution must be made public in the same way because the upstream solution was made public under Article 4(3).

Publication can be on:

- the Interoperable Europe Portal,
- a portal, catalogue or repository connected to the Interoperable Europe Portal.

Operational publication requirements, metadata fields, onboarding processes, and other Portal-specific implementation details may evolve over time and should therefore always be checked against the current guidance and specifications available on the Interoperable Europe Portal.

Legal basis

- **Fulfilment by publication**: The obligation in Article 4(1) may be fulfilled by publishing the relevant content on the Interoperable Europe Portal or a connected portal/catalogue/repository (Article 4(3)).
- **Effect on reuse conditions**: Where the obligation is fulfilled by publication under Article 4(3), Article 4(2) does not apply to the sharing entity (Article 4(3)).
- **Commission publication upon request**: At the request of the sharing entity, the Commission must publish the relevant content on the Interoperable Europe Portal (Article 4(3)).
- **Portal conformance criteria**: Solutions accessible through the Interoperable Europe Portal must:
  - not be subject to third-party rights preventing distribution and use;
  - not contain personal data or confidential information;
  - have a high level of alignment with Interoperable Europe solutions (alignment may be proven by publishing the interoperability assessment outcome);
  - use a licence allowing reuse at least by other Union entities or public sector bodies, or be issued as open source;
  - be regularly maintained under the responsibility of the owner (Article 8(3)(a) to (e)).
- **Publication of adapted solutions**: If an interoperability solution was made public pursuant to Article 4(3), an adapted interoperability solution shall be made public in the same way (Article 4(4)).

3.2.1. Implementation guidance

3.2.1.1. Step 1. Confirm that publication is the appropriate route

> **Article 4(3)**
>
> Article 4(1) may be fulfilled by **publishing** the relevant content on the Interoperable Europe Portal or a connected portal/catalogue/repository.

Publication is the appropriate route where a public organisation wishes to make a solution broadly discoverable and reusable, rather than handling reuse through individual direct-sharing exchanges. In practice, publication may be particularly suitable where:

- reuse by more than one public organisation is expected;
- similar requests are likely to recur;

- the solution is sufficiently stable and documented to be made publicly accessible;
- the public organisation can keep the relevant reference information, links, ownership or contact details, and lifecycle status accurate over time;
- the solution can meet the conditions for conformance with the Interoperable Europe Portal.

Publishing not only serves the public interest by reducing duplication, but can also bring practical benefits such as greater visibility, wider reuse by other public organisations, contributions from a broader community of users or maintainers, and better alignment with Union-level interoperability support actions. It may also contribute to broader digital sovereignty objectives by facilitating inspection, reuse, adaptation and collaborative maintenance of interoperability solutions across public organisations, while reducing dependency on single providers and supporting greater continuity, control and flexibility in the long-term management of public-sector digital solutions.

At this stage, the sharing entity should decide which publication route it intends to use:

- **direct publication on the Interoperable Europe Portal** (including publication by the Commission at the public organisation's request);
- **publication via an existing portal, catalogue or repository** that is connected to the Interoperable Europe Portal.

Both options are valid under Article 4(3). The choice may be guided by practical considerations, including:

- the ability of the organisation to **retain ownership and responsibility** for the published content;
- the capacity to **update and maintain** the published information over time;
- the stability and longevity of **links and identifiers** used.

If publication is not operationally suitable, or if the solution cannot lawfully be made accessible through the Portal, the public organisation may consider the direct-sharing pathway instead, described in Section 3.3.

For open source software solutions, publication through the **EU OSS Catalogue** may**,** where relevant operational conditions are met, provide a practical route. For the purposes of these Guidelines, this is a publication route within the Interoperable Europe Portal ecosystem, not an additional legal route separate from the two options listed above.

Output of this step: a route decision, identifying the chosen publication location.

3.2.1.2. Step 2. Run the Portal-readiness check

Article 8(3)

Solutions **accessible through the Portal** must meet Article 8(3)(a) to (e), including:

- no blocking third-party rights,
- no personal data or confidential information,
- alignment evidence (may be via published interoperability assessment outcome),
- reuse-permitting licence or open source,
- regular maintenance under owner responsibility.

Before publication is initiated, as part of the readiness check, the sharing entity may carry out practical verifications, such as:

- reviewing documentation, examples, test data or sample configurations to ensure that no **personal data or confidential information** is included;
- removing **sensitive information** such as secrets, keys, tokens or internal URLs;
- confirming that **licences** of dependencies and **third-party components** allow distribution and reuse.

If one or more of the conditions for publication cannot be met, the sharing entity should not use the Article 4(3) publication route to fulfil its obligation under Article 4(1). In that case, where Article 4(1) applies, the sharing entity may instead fulfil the obligation through direct sharing under Articles 4(1) and 4(2)), or adjust the publication package so that the interoperability solution can lawfully be made accessible through the Interoperable Europe Portal.

Output of this step: a completed Portal-readiness check, retained by the sharing entity as part of its publication record.

3.2.1.3. Step 3. Prepare the publication package

Article 4(1) & 4(3)

Where Article 4(1) is fulfilled by publication under Article 4(3), the relevant content to be made available should correspond to the content referred to in Article 4(1), namely the interoperability solution, including the technical documentation and, where applicable, the version history, documented source code and the references to open standards or technical specifications used.

The sharing entity prepares a package with the materials to be published, taking into account the nature and complexity of the solution.

Where publication is used to fulfil Article 4(1), **the package should contain the materials needed to make that fulfilment effective**.

Where publication is proactive, the same elements should be used as a practical baseline, to the extent proportionate and available.

Because Article 4(2) governs direct, bilateral sharing on request and does not apply where reuse is enabled through publication under Article 4(3), the public organisation should not frame the publication package around Article 4(2) reuse conditions. If the public organisation wishes to indicate whether support, collaboration, issue reporting or community engagement is available, that information may be included as descriptive information on a voluntary basis.

Early-stage or still-evolving interoperability solutions may also be published where this supports transparency, discoverability, or collaboration, provided that the scope and maturity of the solution are made clear (e.g., in the description). Where publication is used to fulfil Article 4(1) via Article 4(3), the solution still needs to meet the Article 8(3) conditions for being accessible through the Portal.

Output of this step: a publication package associated, where applicable, with a defined version or release.

3.2.1.4. Step 4 Option A. Publish directly on the Interoperable Europe Portal, or request Commission publication

> **Article 4(3)**
>
> At the sharing entity's request, the **Commission must publish** the relevant content on the Interoperable Europe Portal.

When publishing an interoperability solution, the sharing entity may need to provide the **descriptive information required by the Interoperable Europe Portal** to support search, discovery, and understanding by users. Providing complete and accurate metadata also enhances the discoverability of interoperability solutions both within the Portal and through external search engines.

Whether the sharing entity requests the Commission or directly publishes the solution on the Interoperable Europe Portal, it should provide a **complete and coherent publication package** in line with the current Portal workflow to facilitate timely publication and avoid unnecessary back-and-forth.

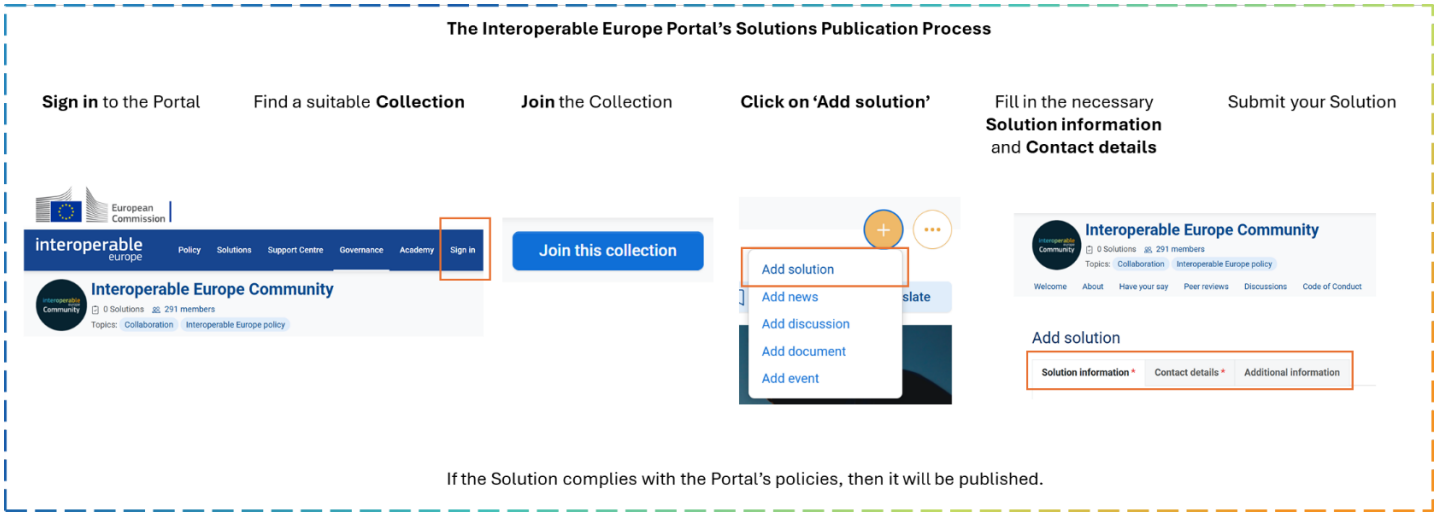
The Commission's role in this context is to support publication on the Portal, whereas responsibility for the accuracy, completeness and maintenance of the content remains with the sharing entity.

Depending on the current interface, the Portal workflow may require descriptive information to support search, discovery and understanding by users. This may include, for example[7]:

- the solution title, abstract and description;
- the solution classification;
- the solution type;
- the domain and topic;
- the language availability;
- the solution owner and contact information.

[7] How to create Solutions on the Portal | Interoperable Europe Portal, How-To Articles | Interoperable Europe Portal, Eligibility criteria | Interoperable Europe Portal

These fields are defined by the Portal as part of its operation and do not constitute legally mandated metadata standards under the Act.



The current Portal guidance also describes optional additional information, including logo, custom URL identifier, related documents, search keywords, related solutions, status, and relevant website. Where readily available, the sharing entity may also indicate the country of origin and any known reuse cases. This information can support findability and relevance assessment.

In focus

The **Interoperable Europe Portal** provides **support resources and guidance** to assist public administrations and solution owners in familiarising themselves with the Portal publication process. These resources, available on the Portal's **Solutions page**, cover key aspects such as roles and responsibilities of solution owners, eligibility criteria, and How-To articles with practical guidance on how to publish solutions. In addition, the Portal team regularly hosts webinars to provide support to users.

The level of detail provided is recommended to remain proportionate and feasible for the publishing entity but complete enough to support discoverability. More complete descriptions may facilitate reuse, but publication does not depend on the provision of extensive metadata beyond what is needed for the chosen route.

Information note: Metadata models and technical implementation

The Interoperable Europe Portal internally relies on structured information models to organise and present solutions. In practice, this includes the use of models such as the Asset Description Metadata Schema Application Profile (ADMS-AP). The Portal also supports machine-readable access to certain information through a SPARQL endpoint. For software solutions linked in the EU OSS Catalogue, alignment with publiccode.yml specifications is used.

Output of this step: a published entry on the Interoperable Europe Portal, or a formal publication request submitted to the Commission, supported by the necessary materials.

3.2.1.5. Step 4 Option B. If publishing on a connected repository, make sure the 'connection' is real and sustainable

> **Article 4(3)**
>
> Article 4(1) may be fulfilled by publishing the relevant content on the Interoperable Europe Portal or a **connected portal/catalogue/repository**.

For the purposes of these Guidelines, this route should be treated as connected only where an operational arrangement ensures that the relevant solution is made accessible through the Interoperable Europe Portal in a stable and maintainable manner. Mere public availability of a repository on the internet does not, by itself, amount to connected publication. The sharing entity may consider the following elements.

- **Stable landing page**

  Each interoperability solution may have a stable landing page. This may be a long-lived URL that remains valid across releases and versions, and that serves as the primary reference point for the solution. Individual versions or releases may be linked from that page, but the landing page itself is recommended to remain stable over time.

- **Persistent identifiers and versioning**

  The solution may be identifiable through a stable identifier, together with explicit version tags or release references. This may help reusing entities to cite precisely which version of the solution they are reusing and supports traceability over time.

- **Predictable update mechanism**

  The sharing entity may decide how updates to the solution will be reflected on the Interoperable Europe Portal. This may include, for example:

  - updating the landing page to point to the latest release;
  - maintaining a visible change log or release history.

  The update approach is recommended to be predictable and documented, so that users are not confronted with broken links or unclear version references. The date of the last update could be stated on the Interoperable Europe Portal.

- **Operation of external catalogues or repositories**

  Where the sharing entity itself operates the external portal, catalogue or repository, it should follow the logic of Article 8(4) by applying necessary and proportionate measures to ensure interoperability with the Interoperable Europe Portal. Technical aspects of portal-to-portal interoperability (including federation mechanisms) are addressed in the connecting to the Interoperable Europe Portal pathway described in Section 3.6.

Output of this step: a stable landing page for the interoperability solution; and a minimal connection plan, describing who is responsible for updates and how broken links or outdated references are avoided.

In focus

The Italian Ministry of Innovation and Technology proposed the **Digital Administration Code** to establish harmonised rules to ensure the availability, management, access and reusability of their IT solutions. Under Article 69, public sector bodies that own IT solutions are required to make the related source code publicly available. The code must be released together with relevant documentation and published in a public repository under an open source license. This ensures that the solution can be easily reused or adapted by other entities to tailor it to their own needs.

3.2.1.6. Step 5. Maintain the published solution

Article 8(3)(e)

Portal-accessible solutions must be **regularly maintained** under the owner's responsibility.

Where a Portal entry or connected landing page is used, maintenance under Article 8(3)(e) requires that the information made accessible through the Portal remain accurate enough for reuse.

This does not require the owner to duplicate detailed technical information across multiple platforms where a single authoritative source can be referenced reliably. In many cases, it will be sufficient to keep the Portal entry or connected landing page aligned with that authoritative source, including current maintenance status, version or release information, and any material change affecting reuse.

The level of effort devoted to upkeep should remain proportionate to the nature of the solution and its reuse. However, information made accessible through the Portal or a connected landing page should remain accurate enough that reusing entities are not misled as to the solution's current status.

Output of this step: a published entry that remains sufficiently up to date over time, with accurate reference information.

In focus

Solution owners can minimise the amount of information maintained directly on the Portal and instead provide **reference to a single authoritative source**, such as the original national repository. This approach reduces the effort required to keep information aligned across multiple locations, lowers the risk that solutions remain outdated or inconsistently maintained on the Portal, and ensures that reusing entities can clearly understand whether a solution is actively maintained, has been updated, or has been superseded.

3.2.2. Monitoring and performance indicators

The IEA does not require the monitoring of specific indicators for publication under Article 4(3). Public organisations may nevertheless choose to monitor the visibility, reuse and impact of published interoperability solutions, applying a proportionate approach.

Basic indicators:

- views or visits to the solution's Portal entry or landing page;
- number of active reusable solutions available through the Portal;
- downloads or access counts for documentation or source code.

Advanced indicators:

- reuse and adoption metrics (e.g. documented reuse cases, deployments, forks);
- community engagement indicators (e.g. issues reported and resolved, external contributions);
- quality indicators (e.g. completeness of metadata, clarity of documentation);
- sustainability indicators (e.g. update frequency, longevity, diversity of contributors).

These indicators are illustrative and should be applied only where proportionate to the nature, maturity and expected reuse of the solution.

3.2.3. Risks and mitigation measures

Publication under Article 4(3) can give rise to legal, technical, organisational and operational risks if the publication package is incomplete, if the solution is made accessible without a sufficient readiness check, or if maintenance responsibilities are unclear.

Typical risks include:

- publication of content affected by third-party rights that prevent lawful distribution and use;
- inadvertent disclosure of personal data, confidential information, or other material that should not be made public;
- unclear licensing or missing notices that make reuse legally uncertain;
- weak versioning or unstable links that make the published solution difficult to identify and cite;
- stale or misleading entries caused by inadequate maintenance;
- confusion between simple connected publication and the more specific interoperability and federation issues dealt with elsewhere in these Guidelines.

These risks can be mitigated by carrying out a structured publication-readiness check before publication, involving legal, technical, security and data-protection functions where appropriate, assigning a clear owner and contact point, using stable references and explicit versioning, and reviewing published entries periodically to ensure that they remain accurate and usable.

3.3. Direct sharing of interoperability solutions upon request (Article 4(1))

This pathway applies when:

- a Union entity or public sector body receives a request from another Union entity or public sector body;
- the sharing entity decides to fulfil the obligation through direct sharing, not by publication under Article 4(3).

Legal basis

- **Obligation to make available upon request**: A Union entity or public sector body must make available to any other Union entity or public sector body that requests it an interoperability solution supporting a trans-European digital public service, including technical documentation and, where applicable, version history, documented source code, and references to open standards or technical specifications used (Article 4(1)).
- **Exceptions**: The Article 4(1) obligation does not apply if the solution:
  - falls outside the scope of the sharing entity's public task (subject to transparency and reviewability);
  - is constrained by third-party intellectual property rights restricting sharing for reuse;
  - is excluded or restricted on specified security grounds (Article 4(1)(a) to (c)).
- **Reuse conditions and support stance**: To enable autonomous management by the reusing entity, the sharing entity must specify any conditions applying to reuse, including any guarantees regarding cooperation, support and maintenance. Those conditions may include exclusion of the sharing entity's liability in case of misuse by the reusing entity (Article 4(2)).
- **Cybersecurity and evolution autonomy assessment**: Before adoption, the reusing entity must, upon request, provide an assessment covering its ability to manage autonomously the cybersecurity and the evolution of the reused interoperability solution (Article 4(2)).

3.3.1. Implementation guidance

3.3.1.1. Step 1. Register and validate the request

> **Article 4(1)**
>
> The obligation applies when a **Union entity or public sector body requests** an interoperability solution supporting a trans-European digital public service.

Requests may be submitted through a published request channel, such as an email address, service mailbox, ticketing system or another official submission channel. Where no direct channel is published, the Member State single point of contact may serve as an operational route. The Act does not impose a specific format.

Guidance for the requesting entity

To facilitate timely handling, a request may indicate:

- **who** is requesting the solution (legal name of the organisation, authority, and a contact point);
- **what** solution is requested (name or identifier, link if publicly known, and scope of interest);
- **why** the solution is needed (which trans-European digital public service is supported, intended reuse, integration or adaptation context);

- **which artefacts** are sought (documentation only, source code, standards references, specific versions);
- **any constraints** relevant to handling the request (indicative timelines, format);
- **any information-handling requirements**, where applicable (for example confidentiality arrangements).

Providing this information does not create additional obligations; it helps reduce back-and-forth exchanges and supports timely sharing.

Guidance for the sharing entity

Upon receipt of a request, the sharing entity may:

- **acknowledge receipt** and indicate next steps;
- **verify** that the requester is a Union entity or public sector body within the scope of Article 4;
- **seek clarification**, where key information is missing or unclear, before proceeding further;
- **log the request** in an appropriate register or ticketing system, where proportionate, including the requester, date, interoperability solution, purpose and intended delivery method.

The Act does not set deadlines for responding to sharing requests. Internal arrangements may therefore determine the handling sequence and channels used.

Where the requester does not fall within the scope of Article 4(1), the request falls **outside the sharing obligation under the Act**. This does not prevent voluntary cooperation under other applicable rules, where lawful.

Output of this step: a registered request with a unique reference identifier.

3.3.1.2. Step 2. Confirm the scope and the solution boundary

Article 4(1)

The obligation concerns an interoperability solution **supporting a trans-European digital public service**.

The sharing entity may confirm what has been requested, the intended mode of reuse, and whether the request can realistically proceed on the basis of the asset as it exists. This practical scoping is intended to identify the requested solution and any obvious constraints relevant to further handling, but it does not replace the formal exception screening under Step 3. In practice, this may include clarifying:

- what **asset** is requested and, where relevant, which **version or release** is of interest;
- whether the request concerns the solution **as a whole or clearly separable components** that already exist in practice;
- whether the requester seeks artefacts for **independent deployment**, access to a centrally operated service, or a limited pilot or test arrangement;
- which **high-level artefacts** are likely to be in scope, such as documentation, source code, specifications, semantic artefacts, configuration materials or other machine-readable components;
- whether effective reuse **depends on** external services, infrastructure, central operations or third-party components that should be identified clearly;

- the intended **operational and organisational context of reuse**, including pilot or production use, governance and ownership assumptions, localisation or language needs, expected user groups, and any dependencies on shared operations.

Where useful, the sharing entity may also indicate any baseline conditions or constraints already apparent and material to the requesting entity's decision whether to proceed, such as:

- the apparent **legal or licensing basis** on which the asset could be shared;
- the **likely mode of reuse**, such as direct transfer of artefacts, controlled access to a centrally operated service, or pilot or test access;
- whether **support, onboarding, hosting or managed-operation arrangements** appear to be unavailable, optional, or likely to require separate arrangements;
- whether the **requested scope** appears broader or narrower than what exists in practice as a reusable asset.

This early clarification can help the requesting entity decide whether to proceed. It should not, however, be used to turn the Article 4(1) obligation into a negotiation process or to delay compliance unnecessarily.

If, following this clarification, the requesting entity no longer wishes to proceed, the request may be closed. If it remains open, the sharing entity should then carry out the formal exception screening under Step 3. Final reuse conditions, support stance and any liability wording are set out under Step 5.

Output of this step: a short scope note, summarising what will be shared, at which version, and for which intended use.

3.3.1.3. Step 3. Carry out exception screening and confirm whether sharing can proceed

Article 4(1) (a) to (c)

The sharing obligation does not apply if one of the Article 4(1) exceptions applies:

- outside public task scope, with transparency and reviewability;
- third-party intellectual property rights restrict sharing for reuse;
- excluded/restricted for specified security grounds:
  - sensitive critical infrastructure protection related information;
  - defence interests, public security, including national critical infrastructure.

Once the request scope is sufficiently clear, the sharing entity should determine whether any of the exceptions in Article 4(1) prevent sharing in whole or in part. Conducting this screening before assembling the shareable package helps avoid unnecessary work where the request cannot lawfully be fulfilled, or where the requesting entity would not wish to proceed on the basis of a partial or otherwise constrained outcome. In practice, the screening may be carried out by addressing the following questions and recording the outcome:

1. **Public task scope**
   Is the requested interoperability solution linked to activities that fall outside the scope of the sharing entity's public task?

   If the sharing entity relies on this exception, the scope of the relevant public task must be transparent and subject to review. The sharing entity may identify the legal basis, binding

rule, or established administrative practice that defines that scope, and indicate how this determination is transparent and subject to review.

This issue may be especially relevant where the requested reuse would, in practice, require the sharing entity to operate or extend a live service for users outside its existing mandate, territorial scope or administrative context. The Act facilitates the sharing and reuse of interoperability solutions, but does not by itself expand the public task, legal competence or service mandate of the entity concerned. Where cross-border service operation or onboarding of external users would go beyond that mandate, the issue should be identified explicitly and assessed under the applicable legal framework.

2. **Third-party intellectual property rights**
   Does the sharing entity hold the rights necessary to share the solution and allow its reuse?

   If not, the sharing entity may indicate, at a practical level, whether the restriction affects the solution as a whole or only certain components, and what this implies for the possibility of sharing for reuse.

3. **Security considerations**
   Would sharing the solution, or parts of it, disclose information to which access is restricted on grounds of critical infrastructure protection, defence interests, or public security?

   If so, the sharing entity may identify whether, and to what extent, the sensitivity prevents or restricts sharing of the solution. An example would be source code, configuration files, or system architecture documentation for a platform used to operate or protect critical infrastructure, where disclosure would reveal sensitive information that could be used to plan disruption or compromise security. Similar considerations may apply to solutions linked to defence systems or public-security operations.

If an exception applies and the request cannot be fulfilled in full or in part, the sharing entity may communicate a **reasoned explanation** to the requesting entity and record the decision for traceability.

Partial sharing where appropriate

Where an exception applies to an interoperability solution as a whole, Article 4 does not require the sharing entity to break it down or to assess, at a granular level, whether individual components could be shared separately. Nor does it require an integrated system to be fragmented into smaller units that do not exist in practice as self-standing interoperability components. Where clearly separable and reusable components do exist such as common building blocks, libraries, vocabularies or standalone modules, public organisations may consider managing them as independent interoperability solutions. Otherwise, the solution may be shared at the level at which it actually exists, subject to the applicable exceptions and to what is needed for meaningful reuse.

If the screening shows that sharing could proceed only in part of the, or only subject to significant constraints, the sharing entity may confirm with the requesting entity whether it still wishes to proceed on that basis before preparing the shareable package.

Output of this step: an exception screening record, indicating the share/cannot share outcome, together with the applicable exception category and a brief justification.

3.3.1.4. Step 4. Assemble the shareable package

> **Article 4(1)**
>
> The solution shall be made available, including:
>
> - technical documentation, and where applicable,
> - version history,
> - documented source code,
> - references to open standards or technical specifications used.

Once the request has been accepted and scoped, the sharing entity should prepare a practical and proportionate package enabling the reusing entity to understand and reuse the solution without unnecessary follow-up exchanges. Where requested materials do not exist or cannot lawfully be shared, this should be stated clearly.

Beyond the legal minimum, the sharing entity may consider adding a small set of orienting materials where these already exists, such as:

- a short **README** explaining what the solution does, its main purpose, and any prerequisites;
- brief **installation or integration notes**, even if high-level;
- a list of **dependencies** and relevant runtime assumptions;
- **a basic architecture overview**, for example a one-page diagram or description.

The Act does not require upgrading, adapting, or improving the solution as part of sharing.

Typical elements of technical documentation

The Act does not prescribe a documentation format or level of detail. In most cases, compliance can be achieved by sharing existing artefacts and public organisations are not expected to produce extensive new documentation solely for the purpose of responding to a request. Where available and relevant, the technical documentation and accompanying information may include:

- system/solution architecture and design (such as diagrams, component descriptions, and data flows);
- source code or documented software components (such as AI models, libraries, and artefacts);
- testing and validation documentation (such as test plans, results, and compliance checks);
- operational documentation (including deployment instructions, configuration, maintenance, and update procedures);
- reuse and integration guides (e.g. user manuals detailing how to adopt, adapt, or integrate the solution, along with licensing information);
- where relevant, a Software Bill of Materials could be included.

These elements are illustrative only. The materials shared should remain proportionate to the complexity of the solution and its intended reuse. Complex systems may require richer documentation, while simple assets (such as a checklist or a small data model) may need only concise explanatory material.

For **non-software assets**, the shareable package may consist primarily of legal documentation, organisational rules, semantic artefacts, data models, interfaces, specifications, governance materials or service-operating documentation, rather than deployable code.

References to standards and specifications

Where the solution relies on standards or specifications, the sharing entity should provide references to those used. Examples include, where relevant:

- technical specifications (including APIs, protocols, data models, semantic models, and security requirements);
- data and semantic standards (such as RDF, OWL, SKOS, XML, JSON-LD, DCAT-AP or SEMIC Core Vocabularies);
- API and interoperability standards (such as REST/HTTP, OpenAPI, OAuth 2.0, OIDC or SAML 2.0);
- security standards and specifications (such as TLS, ISO/IEC 27001 controls or eIDAS-related specifications);
- document and exchange formats (such as PDF/A, CSV, ODF or ASiC containers);
- European interoperability specifications (such as ELIS, eDelivery, eID, eInvoicing (EN 16931), or INSPIRE and related mappings).

Additional considerations:

- if the solution is already accessible (for example via an existing repository), it may be shared by providing a link;
- documentation may be shared in the language in which it is maintained; where feasible, summaries in English can facilitate cross-border reuse;
- artefacts could be provided in machine-readable and accessible formats, to the extent reasonable.

Output of this step: a shareable package identified by a clear version or release tag and ready for delivery to the requesting entity.

In focus

Making **documentation available in multiple languages** is an effective way to facilitate the direct sharing and reuse of an interoperability solution. For instance, the open source software **Oskari** provides full functionality and documentation in English, Finnish and Swedish, with limited functionalities available in nearly a dozen of additional languages. Where needed, **machine translation services**, such as CEF Automated Translation, can be used to further improve access of the documentation.

3.3.1.5. Step 5. Set reuse conditions and support stance

Article 4(2)

For direct sharing, **reuse conditions** must be specified, including any guarantees on cooperation, support, and maintenance. Conditions may include exclusion of liability in case of misuse.

A short reuse conditions statement may be used to describe the terms under which the interoperability solution is made available. It may cover:

- any applicable **legal or licensing conditions** relevant to reuse, including any notices or obligations attached to the materials shared;

- the **mode of reuse**, where relevant, for example independent deployment, controlled access to a centrally operated service, or pilot or test access;
- the **responsibilities of the sharing entity and the reusing entity** as regards operation, support, maintenance, incident handling or contribution back to an upstream solution, where relevant;
- the **support stance**, if any (for example, no support, best-effort support, or limited support, with an indication of the channel where applicable);
- the **maintenance stance** (for example, actively maintained, limited to security fixes, or no longer maintained);
- any **specific constraints or cautions** relevant to reuse (for example, 'do not deploy in production without additional security hardening');
- where appropriate, any **disclaimer or limitation of liability**, aligned with the sharing entity's internal rules and applicable law, including possible limitation or exclusion of liability.

Where the interoperability solution or its documentation is subject to licensing terms, the sharing entity is recommended to provide the relevant licensing information for the materials being made available. Any third-party intellectual property rights or restrictive licensing issues should already have been considered as part of the exception screening described in Step 3. Open source including OSI-compliant licences such as the EUPL may support reusability, but the choice of licence depends on the specific legal and technical context.

Reuse conditions should be expressed clearly and explicitly. Where no support or maintenance is provided, it should be stated.

Article 4(2) does not create an obligation to provide ongoing support, maintenance or enhancements. The requirement is to specify any guarantees that are provided, not to provide guarantees by default. Where additional onboarding, deployment, training, maintenance, support, hosting or managed-operation services are envisaged in connection with reuse, the conditions applicable to those services may be stated clearly and transparently, including any cooperation arrangements and, where relevant, any separate operational responsibilities or cost implications.

The Act does not regulate the financial treatment of such separate operational services. If such arrangements are envisaged under other applicable Union or national law, they should be described clearly and, where relevant, separately from the basic act of making the interoperability solution available. They should not be structured in a way that undermines the effective sharing of the interoperability solution itself. Where Union entities or public sector bodies wish to organise joint evolution of a solution, they may explore separate arrangements, such as those described in the cost-sharing pathway (Section 3.5).

Where necessary (e.g. for handling sensitive but shareable information), public organisations may complement reuse conditions with simple information-handling or confidentiality arrangements. Such arrangements should not delay or make compliance with Article 4(1) conditional.

Output of this step: a reuse conditions statement, shared together with the interoperability solution and stored as part of the request record.

3.3.1.6. Step 6. Decide whether to request an autonomy assessment

> **Article 4(2)**
>
> Before adoption, the reusing entity must, upon request, provide an assessment of its ability to manage autonomously:
>
> - cybersecurity,
> - evolution of the reused solution.

Requesting an autonomy assessment is optional. It is **not a precondition for making the solution available** and should not be used to create unjustified barriers to reuse. In practice, the sharing entity should consider requesting an assessment only where it is proportionate, for example:

- the interoperability solution is **security-sensitive**;
- the interoperability solution is **complex to operate, maintain or patch**;
- the reusing entity intends to **adapt or fork** the solution and diverge from the original implementation.

Where an autonomy assessment is requested, it may remain short, focused and proportionate. The Act does not require any specific format. It may describe the reusing entity's arrangements for:

- who will **own and govern** the reused solution internally;
- how **cybersecurity** will be managed, including patching, vulnerability handling, dependency management and incident response, as well as, when relevant, whether the reusing entity can:
  - identify, assess and remediate vulnerabilities affecting the reused solution;
  - manage dependencies, updates and version changes without undue reliance on the sharing entity;
  - test, approve and deploy security fixes and functional changes;
  - maintain or obtain the technical documentation, source code or interface information needed for secure operation and future evolution;
  - organise incident handling, continuity and governance for the reused solution within its own administrative context.
- how the reusing entity will **maintain, adapt or evolve** the solution over time;
- whether the reusing entity has, or plans to secure, the **technical capacity** needed to operate the solution in its own environment;
- whether **key external dependencies**, integrations or support assumptions are understood.

If an autonomy assessment cannot be provided, or reveals limitations, the sharing entity may consider **appropriate risk-mitigation measures**, particularly in relation to cybersecurity. This does not affect the obligation to make the solution available under Article 4(1).

Output of this step: where requested, an autonomy assessment, received from the reusing entity and recorded as part of the request record.

3.3.1.7. Step 7. Deliver the interoperability solution and accompanying materials

The Act does not prescribe delivery mechanics. The obligation is to make available the content listed in Article 4(1).

The sharing entity may use an **appropriate delivery channel**, considering the sensitivity, size, and format of the materials being shared. This may include, for example:

- access to a code or document repository,
- secure file transfer,
- a controlled download link.

Where the interoperability solution is delivered through a centrally operated or browser-based service, reuse may, where appropriate, consist in granting access rights to the reusing entity, support to onboarding, providing access to a test or pilot environment, or making available the relevant interfaces and administration tools, rather than handing over source code for parallel deployment. In such cases, the conditions governing that form of reuse should be made sufficiently clear to enable the reusing entity to assess its practical, legal and operational implications.

Where the sharing entity has requested an autonomy assessment under Article 4(2), it may clarify whether production onboarding, activation or other go-live steps are sequenced after receipt of that assessment, and which information has already been provided to enable the assessment to be carried out.

The level of security applied may be proportionate to the nature of the solution and consistent with applicable internal policies. Where stronger verification of the recipient or controlled handover is considered necessary, the measures chosen may remain necessary and proportionate and should not create unnecessary barriers to cross-border sharing, in particular where secure remote delivery is feasible.

Output of this step: a delivery confirmation, together with a brief list of the artefacts delivered, recorded as part of the request.

3.3.1.8. Step 8. Close the request and keep a minimal audit trail

Article 4(1)(a)

Where relying on public task scope exception, the scope must be **transparent and reviewable**. This implies that the decision should be capable of being evidenced.

The sharing entity may retain a concise record of how the request was handled. This may include:

- the **original request**,
- the outcome of the **exception screening** (where applicable),
- a reference to the **version or release** of the interoperability solution shared,
- the **reuse conditions statement** provided,
- any **autonomy assessment** received, if one was requested,
- the **delivery confirmation** and list of **artefacts** made available,
- where relevant, a brief **reasoned explanation** for refusal.

Records could be kept in a way that allows the request to be traced and understood if reviewed at a later stage. The level of documentation may remain proportionate to the nature of the solution and the request.

Repeated requests for the same interoperability solution may indicate that publication under the publication pathway could be considered in future.

Output of this step: a request record, confirming that the direct-sharing process has been completed and its outcome documented for internal traceability purposes.

3.3.2. Monitoring and performance indicators

The IEA does not require the monitoring of specific indicators for direct sharing under Article 4(1). However, public organisations may find it useful to monitor how direct sharing functions in practice, in a manner proportionate to their administrative capacity.

Basic indicators:

- number of sharing requests received;
- number of requests fulfilled or declined;
- average time to acknowledge and respond to requests.

Advanced indicators:

- recurring requests for the same interoperability solution (as an indicator for potential publication under Article 4(3) may be more efficient);
- qualitative feedback from reusing entities on clarity, completeness or usability of the shared materials;
- qualitative feedback from sharing entities on clarity and completeness of the autonomy assessment;
- number of cases in which an autonomy assessment was requested and provided.

Any monitoring should remain proportionate to the administrative burden involved and should not delay or condition compliance with Article 4(1).

3.3.3. Risks and mitigation measures

Direct sharing may involve legal, technical, organisational and operational risks if the request is poorly scoped, if exceptions are not screened consistently, if the shareable package is incomplete, or if reuse conditions are unclear.

Typical risks include:

- uncertainty as to whether the requested asset falls within Article 4(1);
- unclear solution boundaries, versioning or dependency assumptions;
- third-party intellectual property restrictions that are discovered too late;
- disclosure of material access to which is excluded or restricted on security grounds;
- vague or implicit reuse conditions that create uncertainty about support, maintenance or liability;
- autonomy assessment requests that are broader than necessary or are handled in a way that delays sharing unnecessarily;
- incomplete documentation that makes practical reuse difficult.

These risks can be mitigated by using a structured intake and scoping process, carrying out a proportionate exception screening early, preparing a clearly versioned package, stating reuse conditions explicitly, limiting autonomy assessment requests to what is necessary under Article 4(2), and keeping a concise but traceable case record. Clear communication between the requesting and sharing entities supports efficient and trustworthy reuse.

3.4. Sharing adapted interoperability solutions (Article 4(4))

This pathway applies when:

- a reusing entity adapts an interoperability solution to its own needs;
- a third party lawfully reuses and adapts an interoperability solution, in cases where reuse and adaptation are permitted under the applicable licence or access conditions;
- there is a question whether and how the adapted solution must be shared.

Related pathways:

- where an adapted interoperability solution is made public under Article 4(3), the publication pathway in these Guidelines applies (Section 3.2);
- where an adapted interoperability solution is made available upon request to another Union entity or public sector body, the direct-sharing pathway applies (Section 3.3);
- where an adapted interoperability solution is accessible through the Interoperable Europe Portal, the conditions set out in Article 8(3) also apply (Section 3.2.1.2).

Legal basis

- **Right to adapt**: A Union entity or public sector body, or a third party reusing an interoperability solution, may adapt it to its own needs unless third-party intellectual property rights restrict adaptation (Article 4(4)).
- **Obligation to publish adapted solution in certain cases**: If the interoperability solution is made public under Article 4(3), the adapted interoperability solution must be made public in the same way (Article 4(4)).

3.4.1. Implementation guidance

3.4.1.1. Step 1. Identify the upstream solution, the version used, and the purpose of adaptation

> **Article 4(4)**
>
> A Union entity or public sector body, or a third party reusing an interoperability solution, may adapt it to its own needs.

The reusing entity may first identify clearly **what is being adapted and why adaptation is needed**. The key reference point is the upstream interoperability solution and the version, release, or other stable reference on which the adaptation is based. Adaptation may concern, for example:

- software components, source code or configurations;
- technical specifications, interfaces or data models;
- semantic artefacts;
- documentation or implementation guidance;
- a mixed package combining several of these elements.

At this stage, the reusing entity may record:

- the name and identifier of the upstream solution;
- the version, release or stable reference used;
- the owner or maintainer of the upstream solution, where known;
- the purpose of the adaptation and the target context in which it will be used;
- the main reason why the upstream solution cannot be reused unchanged.

This initial note may remain proportionate. For a limited adaptation, a short record may be sufficient. For a substantial adaptation, more structured documentation could be useful from the outset.

Output of this step: an adaptation initiation note by the reusing entity.

In focus

CONSUL is an open source participatory democracy platform originally developed by the Madrid City Council in 2015 to support democratic participation processes across public institutions. Designed from the outset to be **reusable and adaptable**, the platform enables reuse at the local, regional and national level.

The **City of Groningen** **reused and adapted CONSUL** to support citizen participation at the local level, tailoring the solution to its specific governance model and objectives. This adaptation illustrates how interoperability solutions can evolve through reuse.

3.4.1.2. Step 2. Check the legal and rights framework for the adaptation

Article 4(4)

Reusing entities may adapt unless third-party intellectual property rights restrict adaptation.

In practice, identifying third-party intellectual rights that may affect the reusing entity's ability to adapt the solution may include checking:

- whether any **third-party code, libraries, components, specifications, documentation or other assets** are embedded in, or necessary for, the solution;
- whether those elements are **subject to rights or terms** that may restrict modification, redistribution or onward publication;
- whether **contractual arrangements** with suppliers or service providers affect the practical ability to adapt or make available the adapted solution;
- whether any new components introduced through the adaptation create additional **licensing or compatibility issues**.

This review should focus on real constraints and does not require a complete intellectual property audit in every case. A proportionate review may often be sufficient. Where relevant, the reusing entity may also record any notice, attribution, licence continuity or compatibility requirements that may apply to the adapted version.

Output of this step: a short rights and constraints note identifying any third-party intellectual property issues or other legal factors relevant to the adaptation and any onward sharing or publication.

3.4.1.3. Step 3. Carry out and document the adaptation in a way that preserves provenance

Not specified as a process requirement in Article 4(4) but good practice for publication.

The reusing entity is recommended to document the adaptation in a way that makes the **relationship between the adapted solution and the upstream solution understandable to future reusing entities**. This is valuable both where publication is mandatory and where it remains optional, because it supports traceability, maintenance, and informed reuse.

In practice, the documentation could cover:

- what was changed and why;
- which upstream version or release was used as the basis for the adaptation;
- whether the adaptation remains closely aligned with the upstream solution or diverges more substantially;
- any significant implications for interoperability, deployment, maintenance, compliance or security;
- the entity responsible for the adapted version.

Versioning and provenance should be clear. Where possible, the reusing entity may use a stable reference, release tag, branch, fork, or equivalent mechanism that allows the adapted solution to be distinguished from the upstream solution.

Reusing entities may use the indicative structure in Annex 6 to document an adapted interoperability solution in a proportionate manner. For the adaptation concerning semantic artefacts, code lists or vocabularies, a lightweight mapping note may be sufficient. At minimum, this may record the source term or code, the target term or code, the type of relationship between them, any transformation rule applied, any terms with no equivalent, and the version and date of the mapping.

Where an adaptation corrects a defect or introduces an improvement of broader value, informing the upstream maintainer or contributing the change back upstream may support sustainability and reduce unnecessary divergence. This is good practice, not a legal obligation under Article 4(4).

Where the adaptation of an interoperability solution forms part of a decision to implement a new interoperability solution or to introduce a substantial change to an existing one, reusing entities should consider whether a mandatory interoperability assessment is required under Article 3 of the Act. This applies irrespectively of whether the adaptation is carried out through a public procurement procedure or by other means. The documentation of adaptations as described above may support such an assessment, where applicable, but does not in itself constitute an interoperability assessment.

Output of this step: documentation on the adaptation of the interoperability solution, with clear provenance and versioning.

3.4.1.4. Step 4. Decide whether publication of the adapted solution is mandatory

Article 4(4)

If the original solution is made public under Article 4(3), the adapted solution must be **made public in the same way**.

The reusing entity should identify how the upstream solution was made available:

- **Upstream made public under Article 4(3)** *(Section 3.2 Publishing interoperability solutions via the Interoperable Europe Portal or connected portals, catalogues, and repositories (Article 4(3)))*

  If the upstream solution was made public via the Interoperable Europe Portal or via a connected portal, catalogue or repository under Article 4(3), the adapted solution must also be made public in the same way.

  'Made public in the same way' refers to the publication route under Article 4(3). It does not require the adapted solution to be hosted in the same repository, catalogue or technical environment as the upstream solution, provided that it is made public through the Interoperable Europe Portal or through a connected portal, catalogue, or repository.

- **Upstream not made public under Article 4(3)** *(Section 3.3 Direct sharing of interoperability solutions upon request (Article 4(1)))*

  If the upstream solution was shared directly under Article 4(1) and was not made public under Article 4(3), Article 4(4) does not impose an obligation to publish the adapted solution. In such cases, publication remains optional and may be considered on a voluntary basis where it supports reuse, transparency or collaboration.

Output of this step: a recorded decision indicating whether publication of the adapted solution is mandatory or optional, with a brief reference to the upstream publication route.

3.4.1.5. Step 5. Prepare the adaptation package for publication or sharing

Publication route is mandated only in the Article 4(3) upstream case.

As good practice, the adaptation package should enable another reusing entity to understand the adapted solution, its relationship to the upstream solution, and any significant consequences of the adaptation. In practice, it may include:

- the **title or identifier** of the adapted solution;
- a clear **reference** to the upstream solution and version or release used;
- a short **change log** (what has been changed and why);
- sufficient **technical documentation** to understand and assess the adapted solution;
- where relevant, a **semantic mapping** or correspondence table showing how code lists, vocabularies, concepts or data fields have been aligned;
- where applicable, **version history, release identifiers, documented source code, and references to open standards or technical specifications used**;
- the applicable **licence** or other relevant rights information for the adapted solution;
- the **identity of the owner or maintainer** of the adapted solution and a contact point, where appropriate;
- any **significant limitations, dependencies, or risks** that a future reuser should know.

The level of detail may remain proportionate to the complexity of the adaptation and sufficient to allow informed reuse without additional clarification. Where the adaptation is limited and remains very close to the upstream solution, the reusing entity should, where possible, rely on references or links to upstream materials instead of duplicating them, provided that the adapted elements and their effects are still clearly identified.

Output of this step: a package ready for publication or direct sharing of the adapted solution, including the information necessary to understand its relationship to the upstream solution.

3.4.1.6. Step 6. Publish or choose the appropriate sharing route

Article 4(4)

Publish the adaptation in **the same way** where the upstream was made public under Article 4(3).

If publication is mandatory under Article 4(4)

If Step 4 shows that publication is mandatory, the reusing entity should follow the publication pathway in these Guidelines, described in Section 3.2.

Publication may make the provenance of the adapted solution clear by indicating, at a minimum:

- the upstream solution and the version or release on which the adaptation is based;
- the identity of the adapting entity or other responsible entity;
- the version, release or stable reference of the adapted solution.

When the adapted solution is made accessible through the Interoperable Europe Portal, the reusing entity should also ensure that the Article 8(3) conditions are met.

If publication is optional

Where Step 4 shows that publication is not mandatory, the reusing entity may decide whether the adapted solution will:

- remain internal for the time being;
- be published voluntarily under the publication pathway; or
- be shared directly if and when another Union entity or public sector body requests it (Section 3.3).

If the public organisation decides to make the adapted solution available to others, it should use either the publication pathway or the direct-sharing pathway, as appropriate, rather than creating an informal route outside the structure of the Act.

Output of this step: a published adapted solution or recorded decision.

3.4.1.7. Step 7. Manage maintenance, upstream alignment, and future evolution

Article 8(3)(e)

The interoperability solution accessible through the Interoperable Europe Portal shall be regularly maintained under the responsibility of the owner.

The **owner of the upstream solution is not responsible for maintaining adaptations** developed by reusing entities.

To **support sustainability and avoid unnecessary fragmentation**, the reusing entity could define, in a proportionate manner:

- who owns and maintains the adapted solution;
- whether the adapted solution is intended to track upstream updates closely or to diverge over time;
- how upstream security patches, bug fixes and functional improvements will be monitored and assessed;
- how changes to licensing, dependencies, or compliance requirements will be handled;
- whether contributions back to the upstream solution are envisaged;
- how the lifecycle status of the adapted solution will be communicated to future reusers.

This does not require the adapted solution to remain permanently aligned with the upstream solution.

Output of this step: a short maintenance and evolution note for the adapted solution.

3.4.2. Monitoring and performance indicators

The IEA does not require the monitoring of specific indicators for adapted interoperability solutions under Article 4(4). Reusing entities may, however, monitor the health and sustainability of adapted solutions where this supports responsible reuse.

Basic indicators:

- whether the adapted solution clearly identifies the upstream solution and its version;
- whether the publication obligation was met where the upstream solution was made public under Article 4(3);
- whether critical upstream updates or security patches are assessed and, where relevant, incorporated in a timely manner.

Advanced indicators:

- degree of divergence from the upstream solution over time, where this is relevant to sustainability;
- number of contributions or fixes proposed to the upstream solution;
- documented cases of reuse of the adapted solution by other public organisations.

Any monitoring should remain proportionate and should not discourage adaptation or reuse.

3.4.3. Risks and mitigation measures

Adaptation can create legal, technical, organisational and operational risks if provenance is unclear, if the publication trigger in Article 4(4) is missed, or if the adapted solution diverges from the upstream solution without adequate governance.

Typical risks include:

- failure to identify third-party intellectual property constraints affecting adaptation or onward publication;
- unclear documentation of what changed and which upstream version was used;
- loss of interoperability caused by unmanaged divergence from the upstream solution;
- uncertainty about whether publication of the adapted solution is mandatory;
- publication of an adapted solution without meeting the conditions applicable under the publication pathway;
- unclear ownership and maintenance responsibility for the adapted version;

- failure to monitor relevant upstream updates, including security fixes.

These risks can be mitigated by checking the rights framework early, recording the upstream reference and the main changes made, making an explicit decision on whether publication is mandatory, following the publication pathway where Article 4(4) requires publication, assigning a clear owner to the adapted solution, and defining how upstream changes and long-term divergence will be managed.

3.5. Cost-sharing agreements for future developments of interoperability solutions (Article 4(5))

This pathway applies when sharing and reusing entities **decide to jointly fund future developments** of an interoperability solution, such as new features, scaling, hardening or localisation.

Legal basis

- **Permitted cost-sharing**: Sharing and reusing entities may conclude an agreement on sharing the costs for future developments of the interoperability solution (Article 4(5)).

3.5.1. Implementation guidance

3.5.1.1. Step 1. Confirm that funding is for 'future developments', not basic sharing

> **Article 4(5)**
>
> Sharing and reusing entities may conclude an agreement on sharing the costs for **future developments** of the solution.

Article 4(5) is distinct from the obligations in Article 4(1) to 4(3). It allows participating entities to agree voluntarily on cost-sharing for future developments, that is, enhancements or extensions going beyond the interoperability solution already made available under the applicable sharing or publication route.

Such arrangements may help reduce duplication, support coherent evolution of the solution, and enable participating entities to invest jointly in improvements of common value. They do not affect access to an interoperability solution that is already shareable or publishable under the Act.

The Act does not require entities to co-fund future developments, nor does it create a right to request financial contributions from others. The design, scope and terms of such voluntary cooperation remain entirely at the discretion of the participating entities.

Typical situations under Article 4(5) include:

- reuse of an existing interoperability solution by another public body, followed by a voluntary agreement to co-finance **future enhancements or developments**;
- coordinated development roadmaps where a reusing entity contributes financially to **new features** that also benefit the original solution;
- joint funding of **specific future modules or extensions** of a shared solution, without affecting the obligation to share the existing solution;
- an organisation reusing an interoperability solution but requiring additional features decides to enter into a **procurement agreement with the original developer**, with all improvements contributed back to the upstream project.

For the purposes of Article 4(5), the arrangement is between the participating entities. Separate procurement, hosting, development or other contractual arrangements with private actors may accompany it, subject to other applicable law, but they do not alter the scope of Article 4(5) itself. Where the parties rely on a cooperation or administrative arrangement rather than a procurement procedure, they may record, in a proportionate manner, the legal basis and governance rationale for that choice. These Guidelines do not determine that classification.

Routine operational service costs, onboarding of users, hosting, transaction-based usage or support activities are not, as such, 'future developments' within the meaning of Article 4(5). They should not be treated as cost-sharing under that provision unless they relate to agreed future enhancements, extensions or other development work on the solution.

At this stage, participating entities may distinguish clearly between:

- the **baseline interoperability solution** already shared, published or otherwise made available; and
- the **future developments** for which cost-sharing is being considered.

Participating entities are encouraged to identify potential partners by reaching out to entities with similar needs, **across the Union, Member States**, and **administrative levels**.

Output of this step: a scope statement specifying what future developments are envisaged and potentially subject to cost-sharing.

In focus

OS2borgerPC and **MedborgarPC** provide an example of how shared financing with outsourced development functions in practice. Danish and Swedish local government associations (OS2 and Sambruk) jointly fund a common open source codebase, while the actual **software development** and **maintenance** are **outsourced to Magenta**, a Danish IT supplier.

3.5.1.2. Step 2. Agree the core terms of the future-development cost-sharing agreement

The Act permits agreements; it does not prescribe content.

Participating entities may formalise cooperation through an agreement that sets out the scope of the future developments, how they will be funded, and how decisions will be taken. Additional clauses may be included where proportionate, but Article 4(5) does not require a fully developed collaboration framework in every case. Clear communication and transparency from the outset are essential to avoid misunderstandings and ensure balanced collaboration.

To support clearer understanding across Member States, it is recommended, where feasible, to provide **official translations of both the cost-sharing agreement and the documentation of the co-developed solution**. This practice may reduce legal and linguistic ambiguities and ensures that all participating entities operate with accurate and consistent documentation.

Model cost-sharing agreement table of contents

Such agreements typically set out the following elements, as relevant and proportional:

1. Participating sharing and reusing entities
2. Background and purpose
3. Definitions
4. Scope of the cost-sharing agreement *(i.e. future developments covered)*
5. Cost categories and Total Cost of Ownership
6. Cost-allocation model *(e.g. equal shares, population-based, user-based, hybrid)*
7. Payment terms
8. Accounting, reporting, and audit
9. Governance and decision-making

10. Change management
11. Intellectual property and licensing
12. Confidentiality and information handling
13. Legal compliance
14. Duration and termination
15. Liability and risk allocation
16. Dispute resolution

Where proportionate, the cost-sharing agreement may also clarify how enhancements are prioritised, how new partners may join, and how existing partners may withdraw. The participating public organisations may also clarify, to the extent permitted by the applicable legal framework, the applicable law, dispute-handling mechanism, liability allocation, escalation path, and rules on termination or withdrawal. The Act allows such cooperation but does not harmonise these matters.

Define the work packages and deliverables

Participating entities may organise the future developments into work packages with clear deliverables and acceptance criteria:

- feature/module,
- security hardening,
- documentation (including interoperability assessments when introducing new binding requirements that affect cross-border interoperability),
- interoperability extension,
- localisation.

Entities are encouraged to design cost-sharing agreements around modular components, so that to ensure that collaboration remains open, flexible, and attractive to future partners.

It is recommended that participating entities have access to the relevant repositories or shared development environments (or other collaboration tools), where this may support delivery, maintenance or transparency of progress.

Set governance and decision rules

Governance arrangements should remain proportionate to the scale of the future developments. In simpler cases, a short written agreement on prioritisation, decision-making and validation may be sufficient.

Participating entities are recommended to agree on clear governance arrangements to support effective decision-making, coordination, and the long-term sustainability of the collaboration. This may include defining:

- who is responsible for prioritisation;
- how decisions are taken (e.g. consensus, majority);
- how coordination between participating entities is organised (e.g. through a steering group or designated coordination role)
- how scope changes are reviewed and approved;
- how releases are validated.

Depending on the legal and operational context, participating entities may choose, for example, a lead-administration model, a joint steering model, a separately procured development model, or, where legally feasible, stewardship through a neutral non-profit or foundation. These examples are illustrative only.

Governance arrangements may also address continuity over time, including onboarding of new participants, handling of withdrawal of existing participants, and arrangements for maintaining the solution beyond the initial development phase. Participating entities may prepare an accessible 'way of working' document that outlines the tasks and responsibilities each group involved, as well as the process for escalation and conflict resolution.

Cost-sharing can encourage reuse by allowing new participants to join at a later stage to contribute to future developments without reopening or reallocating costs incurred before their participation.

Define the cost allocation and payment method

Participating entities can define a clear and workable method for allocating costs linked to the agreed future developments. Allocation models may include:

- equal shares,
- usage-based contributions,
- capability-based contributions,
- population-based measures,
- hybrid approaches,
- contributions tied to specific work packages.

Where useful, a shared understanding of **total-cost-of-ownership** (TCO) can support realistic planning of long-term costs, which are frequently hidden or underestimated. TCO models in cost-sharing settings often consider the following categories:

- acquisition and set-up (evaluation, solution design, initial configuration and integration);
- operations and management (hosting, monitoring, documentation updates, user support);
- enhancements and upgrades;
- security and resilience (vulnerability management, incident handling, secure development practices);
- training and adoption support (onboarding, guidance, community engagement);
- corrective and adaptive maintenance;
- coordination and governance (steering, release planning, quality assurance);
- end-of-life, exit, and migration planning, where relevant.

Cost-sharing under Article 4(5) may include not only financial contributions but also **in-kind contributions**. In practice, participating entities may contribute through:

- allocation of developers, testers or architects to a joint project team;
- provision of infrastructure, hosting, or tooling;
- translation, documentation, or semantic modelling expertise;
- security review or quality assurance capacity.

Where such in-kind contributions replace or complement financial transfers, the agreement is advised to define:

- the expected scope of work and deliverables,
- the estimated value or effort (for example, in person-days or working hours),
- reporting and accountability arrangements,
- intellectual property and licensing implications of contributed work.

In-kind contributions may be reflected transparently in governance and decision-making arrangements to ensure fairness and sustainability among participants.

In focus

Below are examples of practices illustrating different cost-sharing approaches.

- ZenDiS participated in several joint-investment models. An example is the collaboration with France and the Netherlands on **OpenDesk**, an open source workspace suite for the public sector. Within this **cross-border collaboration**, the involved countries jointly funded the development of core libraries, such as the collaborative writing and note-taking application Docs, which has been integrated in OpenDesk.
- German open source digital integration platform **Integreat** has been deployed across public administrations in Germany under collaborative agreements. Participating municipalities contribute financially by paying a **service fee that is proportionate** to their population size and covers implementation support. The resulting revenues are reinvested in technical maintenance, further development, user support and scaling activities, thereby contributing to the long-term sustainability of the solution.
- **iMio** is a Belgian non-governmental organisation established to coordinate shared solutions and services across municipalities. **Municipalities contribute** financially **based on the specific services they use.** This usage-model allows smaller municipalities to procure and develop IT solutions in line with their financial capacity, paying only for the services they consume.
- Finnish **Digitransit** ensures financial sustainability through a joint **annual budget shared equally** by three public entities. Each partner contributes one third of the total cost, which supports the core development of the solution. However, the development of additional features requires supplementary funding.
- Financial support from organisation benefiting from the solutions may also be regarded as a form of cost participation. **OS2** in Denmark has implemented a framework that **combines more than one allocation model**: first, the members pay an annual membership fee that depends on the size of the municipality, which helps supporting the work of the Secretariat and its employees; then, municipalities contribute to the projects they are interested in and pay an additional 9% fee.

Lock in licensing and publication approach for the outputs

Where future developments are made accessible through the Interoperable Europe Portal, they must meet the conditions set out in Article 8(3), described in Section 3.2. Therefore, the agreement is advised to set out:

- the licence applied to new deliverables;
- whether, when, and by whom outputs will be published on the Interoperable Europe Portal or a connected repository;
- how adaptations or evolutions will be handled so that publication obligations are met.

Clear provisions on ownership, licensing, and publication responsibilities help ensure that jointly developed components remain reusable, accessible and aligned with the interests of all parties involved.

Ensure administrative feasibility

Because the Act allows cost-sharing agreements but does not harmonise national administrative rules, each participating entity may **confirm that the agreement is feasible under its own legal, organisational, and financial framework**. Entities may find it useful to verify:

- an appropriate procurement route is available under national law;
- budget cycles and internal approvals are aligned with the planned contributions;

- audit and reporting requirements can be met;
- any other national rule is respected;
- the agreement sufficiently addresses applicable law, dispute handling, liability and termination arrangements, where relevant.

Output of this step: a signed (or otherwise formally approved) cost-sharing agreement describing scope, governance, cost allocation, and licensing and publication responsibilities.

3.5.1.3. Step 3. Deliver, publish where relevant, and review

Where outputs are made public to fulfil the sharing obligation under Article 4(1) via publication under Article 4(3), adaptations and evolutions of the solution are subject to the publication obligation in Article 4(4) and the access conditions set out in Article 8(3).

Once the cost-sharing arrangement is in place, participating entities implement the agreed work packages and manage delivery in line with the agreed governance arrangements.

Where outputs are later made available under one of the pathways in Chapter **Error! Reference source not found.**, the relevant pathway may be followed. In particular:

- if outputs are published under Article 4(3), the publication pathway may be used (Section 3.2);
- if outputs are shared upon request, the direct-sharing pathway may be used (Section 3.3);
- if jointly funded work results in an adapted solution, the adaptation pathway may be relevant (Section 3.4).

After delivery and, where relevant, publication, a structured review process may help maintain transparency, track progress, and consider that the cooperation remains aligned with shared goals. Because requirements may evolve rapidly, participating entities are recommended to **conduct periodic reviews** covering:

- delivery status of agreed work packages,
- expenditures compared with planned contributions,
- onboarding of new users (where relevant),
- whether the cooperation should be continued, adjusted, or renewed.

These reviews may also involve updating information on the Interoperable Europe Portal or a connected repository when new versions, improved documentation, or compatibility changes become available. Adjustments can be made through negotiations, arrangements, or the use of flexible contractual mechanisms.

Output of this step: release notes and review record documenting progress, publication, and any other relevant decision.

3.5.2. Monitoring and performance indicators

The IEA does not prescribe specific indicators for cost-sharing agreements under Article 4(5). Participating entities may nevertheless monitor financial, organisational, and technical aspects of cost-sharing arrangements to support informed decision-making.

Basic indicators:

- execution of agreed financial contributions;
- delivery of planned work packages or enhancements.

Advanced indicators:

- uptake of the solution by additional public organisations;
- sustainability of maintenance arrangements over time;
- alignment between planned and actual costs for future developments.

Any monitoring approach should remain proportionate and adaptable to the scale, duration and complexity of the shared solution.

3.5.3. Risks and mitigation measures

Cost-sharing arrangements may give rise to legal, financial, technical, and organisational risks if the baseline solution and the future developments are not clearly distinguished, if contribution models are unclear, or if governance and publication responsibilities are left implicit.

Typical risks include:

- using cost-sharing in a way that appears to condition access to the baseline solution that should already be shareable or publishable under the applicable pathway;
- uncertainty about which costs are in scope and which are not;
- unbalanced contribution models that undermine fairness or long-term participation;
- insufficient governance, leading to prioritisation disputes, delivery delays, or deadlock;
- unclear ownership, licensing, or dependency management for jointly funded outputs;
- failure to identify when outputs should be published or maintained under the applicable pathway;
- misalignment between the agreement and the participants' internal budgetary, procurement, or audit rules.

These risks can be mitigated by distinguishing clearly between the baseline solution and future developments at the outset, defining work packages and deliverables precisely, using a transparent cost-allocation model, assigning governance and publication responsibilities explicitly, checking administrative feasibility before signature, and reviewing the arrangement periodically so that issues can be addressed before they undermine the cooperation.

3.6. Connecting portals, catalogues, and repositories to the Interoperable Europe Portal (Article 8(4))

This pathway applies when:

- a Union entity or public sector body provides a portal, catalogue, or repository with similar functions to the Interoperable Europe Portal;
- Article 8(4) requires it to ensure interoperability with the Interoperable Europe Portal through necessary and proportionate measures.

Legal basis

- **Interoperability obligation**: Where a Union entity or public sector body provides a portal, catalogue or repository with similar functions, it must take the necessary and proportionate measures to ensure interoperability with the Interoperable Europe Portal (Article 8(4)).
- **EUPL support in open source portals**: Where such portals collect open source solutions, they must allow for the use of the European Union Public Licence (Article 8(4)).

3.6.1. Implementation guidance

3.6.1.1. Step 1. Confirm whether the platform is in scope

> **Article 8(4)**
>
> Where a Union entity or public sector body provides a portal/catalogue/repository with similar functions, it must take necessary and proportionate measures to ensure **interoperability with the Interoperable Europe Portal**.

To support the assessment of whether a platform is in scope of the Article 8(4), it may focus on the platform's actual function. A platform is more likely to be in scope where it is intended to organise, describe, make discoverable, or provide access to interoperability solutions for reuse by actors beyond a single internal team. It is recommended to answer and document the following elements:

- **function of the platform**: is it primarily a catalogue, a repository, or a combination of both?
- **type of content hosted:** does the platform host interoperability solutions?
- **intended audience and reuse purpose:** is the platform intended to support reuse by other organisations and/or attract contributors?
- **content coverage:** which content would be made accessible through the Interoperable Europe Portal (all entries or a curated subset)?

Output of this step: a short scope note by the platform owner confirming whether the platform is considered in scope of Article 8(4) and the reasoning supporting that conclusion.

3.6.1.2. Step 2. Determine the appropriate interoperability approach

Article 8(4)

Measures must be **necessary and proportionate**.

Before integrating with the Interoperable Europe Portal, the **level and form of interoperability appropriate** for platform in question is to be clarified. Article 8(4) requires the necessary and proportionate measures to ensure interoperability with the Interoperable Europe Portal; it does not prescribe a single integration model. At an early stage, the entity may consider:

- the size, maturity and complexity of the platform;
- the number and frequency of updates to the solutions concerned;
- the expected level of cross-border visibility and reuse;
- the resources and technical capacity available;
- whether the content includes open source software and whether federation with the EU OSS Catalogue is envisaged.

In practice, interoperability may range from simple visibility and linking to more structured metadata exchange or more automated forms of federation. Depending on the size, maturity, and purpose of the platform, this may include:

- **linking and manual curation**: each solution has a stable landing page or other durable public reference, and selected entries are manually uploaded, referred or otherwise curated for visibility through the Portal;
- **federation**: a structured and largely automated process in which a catalogue exposes structured metadata for harvesting or synchronisation. In the case of EU OSS Catalogue this relies on a public API and publiccode.yml.

The entity may also decide whether the Interoperable Europe Portal will expose:

- all relevant entries on the platform; or
- a curated subset of entries that meet the chosen criteria and can be maintained reliably over time.

This decision may be documented in a proportionate manner. The purpose is not to impose an unnecessarily complex solution, but to ensure that the level of interoperability chosen is real, sustainable and justified.

Output of this step: an interoperability objective note identifying the content scope, the intended level of interoperability, and the reasons why that level is considered necessary and proportionate.

3.6.1.3. Step 3. Check which entries can be exposed through the Interoperable Europe Portal

Article 8(3)

Solutions **accessible through the Portal** must meet Article 8(3)(a) to (e), including:

- no blocking third-party rights,
- no personal data or confidential information,
- alignment evidence (may be via published interoperability assessment outcome),
- reuse-permitting licence or open source,
- regular maintenance under owner responsibility.

The entity should check whether the entries intended to be exposed through the connection can be made accessible through the Interoperable Europe Portal in compliance with the relevant conditions. This check should remain proportionate. For a curated subset of mature entries, it may be straightforward. For a larger platform with mixed content, the entity may apply the check selectively and expose only the subset that can meet the conditions.

The **review should also cover the metadata and exchange layer**. APIs, feeds, landing pages and exported records should not disclose personal data, confidential information, secrets, or other material that should not be made publicly accessible.

If some entries do not meet the Article 8(3) conditions, they cannot be made accessible through the Interoperable Europe Portal nor through the EU OSS Catalogue under this route. The entity may then:

- exclude those entries from the interoperable subset;
- adjust the exposed materials so that the conditions can be met; or
- reconsider the scope of the connection.

Output of this step: an Interoperable Europe Portal-conformance check confirming which entries can be made available through the Interoperable Europe Portal and on what basis.

3.6.1.4. Step 4. Establish baseline content, identifiers and descriptive information

This step has an indirect legal basis in Article 8(4), and the content of the step is operational good practice.

Before implementing a technical connection, the platform owner may check that the content to be exposed through the Interoperable Europe Portal is sufficiently stable and intelligible. Regardless of the technical model chosen, **each exposed entry** could have:

- a stable landing page or equivalent durable public reference;
- a clear title and concise description;
- an identifiable owner or responsible organisation;
- licence information, where relevant;
- version or release information, where relevant;
- a clear indication of lifecycle or maintenance status, where available;
- links to the relevant repository, documentation, distribution, or other source materials.

Stable identifiers and explicit versioning may be particularly useful where solutions evolve over time. They support discoverability, traceability and reliable citation across platforms.

The platform owner may also define how updates, deprecations, archiving and withdrawals will be reflected, so that users are not confronted with broken links, unclear version references, or ambiguous solution status.

At this stage, the platform owner may identify the core descriptive fields used locally and map them, in a proportionate way, to the information needed for cross-platform discoverability. The Act does not impose a single metadata schema for this purpose. However, as the level of automation increases, structured and machine-readable metadata may become increasingly useful in practice.

Output of this step: a baseline interoperability inventory for the entries of the platform to be exposed on the Interoperable Europe Portal.

3.6.1.5. Step 5. Implement the chosen connection model

Article 8(4)

Interoperability should be implemented through necessary and proportionate measures that make the relevant solutions accessible through the Interoperable Europe Portal.

The platform owner may implement the connection model selected in Step 2 in a way that is maintainable over time. Whichever model is chosen, it could **document the chosen approach** and identify who is responsible for operating and maintaining it.

Different types of assets are best described using different metadata profiles. Selecting an approach that matches the content of the platform helps ensure interoperability while avoiding unnecessary complexity. Commonly used options are presented below.

DCAT-AP

Designed for the exchange of dataset and data service catalogues in Europe and maintained under SEMIC. This profile is particularly suitable where a platform primarily functions as a data catalogue.

ADMS/ADMS-AP

Designed to describe interoperability assets and solutions and used within the Interoperable Europe ecosystem for solution description and federation workflows.

CPSV-AP

Useful where a platform also describes public services in a structured way and where making that service layer machine-readable is relevant.

Publiccode.yml

A repository-level metadata format for public sector software, used by the EU OSS Catalogue as its reference model for discovering and filtering open source solutions.

The use of open and commonly adopted data models facilitates exchange and reuse. In practice, metadata is often expressed using RDF-based serialisations (such as JSON-LD or Turtle) and widely used vocabularies (for example, SKOS or Dublin Core), or, in the case of software repositories, using formats such as YAML for publiccode.yml.

Reference-level or structured interoperability

Where a lighter or intermediate model is proportionate, interoperability may be achieved through measures such as:

- stable public landing pages for each entry;

- clear and consistent descriptive information;
- manual or periodic synchronisation of selected entries;
- machine-readable exports or feeds containing a defined subset of descriptive fields.

This model may be appropriate where the number of entries is limited, updates are infrequent, or the platform is at an early stage of maturity.

Federation of open source catalogues with the EU OSS Catalogue

Where the sharing entity collects open source software solutions and seeks integration with the EU OSS Catalogue, the current operational model is more specific. Automated federation is based on conditions including:

- public git repositories for the solutions concerned;
- a valid publiccode.yml file in each repository;
- a publicly accessible API that allows the source catalogue or solution data to be retrieved and synchronised.

The EU OSS Catalogue uses publiccode.yml as its principal metadata specification for this purpose. These are operational onboarding and federation requirements of that catalogue model. They should therefore be treated as current practical conditions for that specific integration route, not as general legal requirements of Article 8(4) for all portals, catalogues or repositories.

Output of this step: a working connection mechanism, supported by a short technical note explaining the chosen interoperability model, the exposed content scope, and the maintenance approach.

In focus

The **EU OSS Catalogue** exemplifies effective **integration between repositories**. It provides a curated selection of open source solutions, consolidating information on software packages and their use by European public administrations. Its value lies in aggregating data from multiple sources and presenting it in a standardised format. The catalogue is built on a **federated architecture of public APIs**, creating a central pool of open source solutions sourced from national, regional, and local open source software catalogues across EU Member States. As part of the Interoperable Europe Portal, the provisions of Article 8(3) apply.

3.6.1.6. Step 6. Where open source solutions are collected, ensure that the use of the EUPL is not excluded

Article 8(4)

Where such portals collect open source solutions, they must allow the use of the **European Union Public Licence** (EUPL).

This obligation is best understood as a **non-exclusion requirement**. It means that the platform should not block, refuse, mis-handle, or otherwise prevent the use of the EUPL in practice. It does not require the EUPL to be the default licence, and it does not prevent the platform from also allowing other open source licences. In practice, the entity may verify, where relevant, that:

- the EUPL can be selected, declared or recorded wherever licence information is captured;
- licence fields, metadata models, validation rules and onboarding workflows do not reject or distort EUPL-licensed entries;

- landing pages and machine-readable outputs can display or transmit EUPL information clearly and consistently;
- free-text or non-standard licence handling does not create ambiguity that effectively disadvantages EUPL-licensed solutions.

Where the platform does not itself assign licences but merely exposes externally maintained entries, the entity may still ensure that its platform logic and metadata handling can accommodate the EUPL without obstruction.

Output of this step: a record confirming that EUPL-licensed entries are not excluded by the platform.

3.6.1.7. Step 7. Test end-to-end, assign responsibilities, and maintain interoperability over time

> **Article 8(4)**
>
> Interoperability with the Interoperable Europe Portal should not be treated as complete once the initial connection is created. The measures required by Article 8(4) remain effective only if the connection continues to function over time.

Before go-live, the platform owner may **carry out an end-to-end testing on a small and manageable subset of entries**. Depending on the model used, this may include checking that:

- entries can be discovered through the Interoperable Europe Portal;
- links resolve reliably to the intended landing pages;
- licence, version and ownership information remains visible and consistent;
- machine-readable exports, feeds or APIs are accessible and behave as expected;
- duplicate, withdrawn or deprecated entries are handled in a predictable way.

The entity may also assign responsibilities for:

- technical maintenance of the connection;
- metadata quality and update handling;
- issue resolution and contact with the Interoperable Europe Portal team;
- lifecycle management for deprecated or withdrawn entries;
- review of schema, mapping or API changes that may affect interoperability.

A short operational governance note is usually sufficient. The main objective is that roles and update routines are clear enough to prevent silent degradation of interoperability over time.

Where exposed entries remain accessible through the Interoperable Europe Portal, the solutions themselves must be regularly maintained under the responsibility of their owner in accordance with Article 8(3).

Output of this step: a short test report and an operational governance note describing who maintains the connection, how updates are handled, and how interoperability issues are identified and resolved over time.

3.6.2. Monitoring and performance indicators

The IEA does not require the monitoring of performance indicators for interoperability between portals or repositories under Article 8(4). Public organisations may nevertheless track a small number of indicators to assess whether integration measures function as intended.

Basic indicators:

- number of portals, catalogues or repositories successfully integrated with the Interoperable Europe Portal;
- number of entries from portals, catalogues or repositories exposed by the Interoperable Europe Portal;
- stability of links to external solution landing pages.

Advanced indicators:

- average time required to complete an integration process;
- frequency and timeliness of metadata or content updates.

Any monitoring should remain necessary and proportionate to the level of integration implemented.

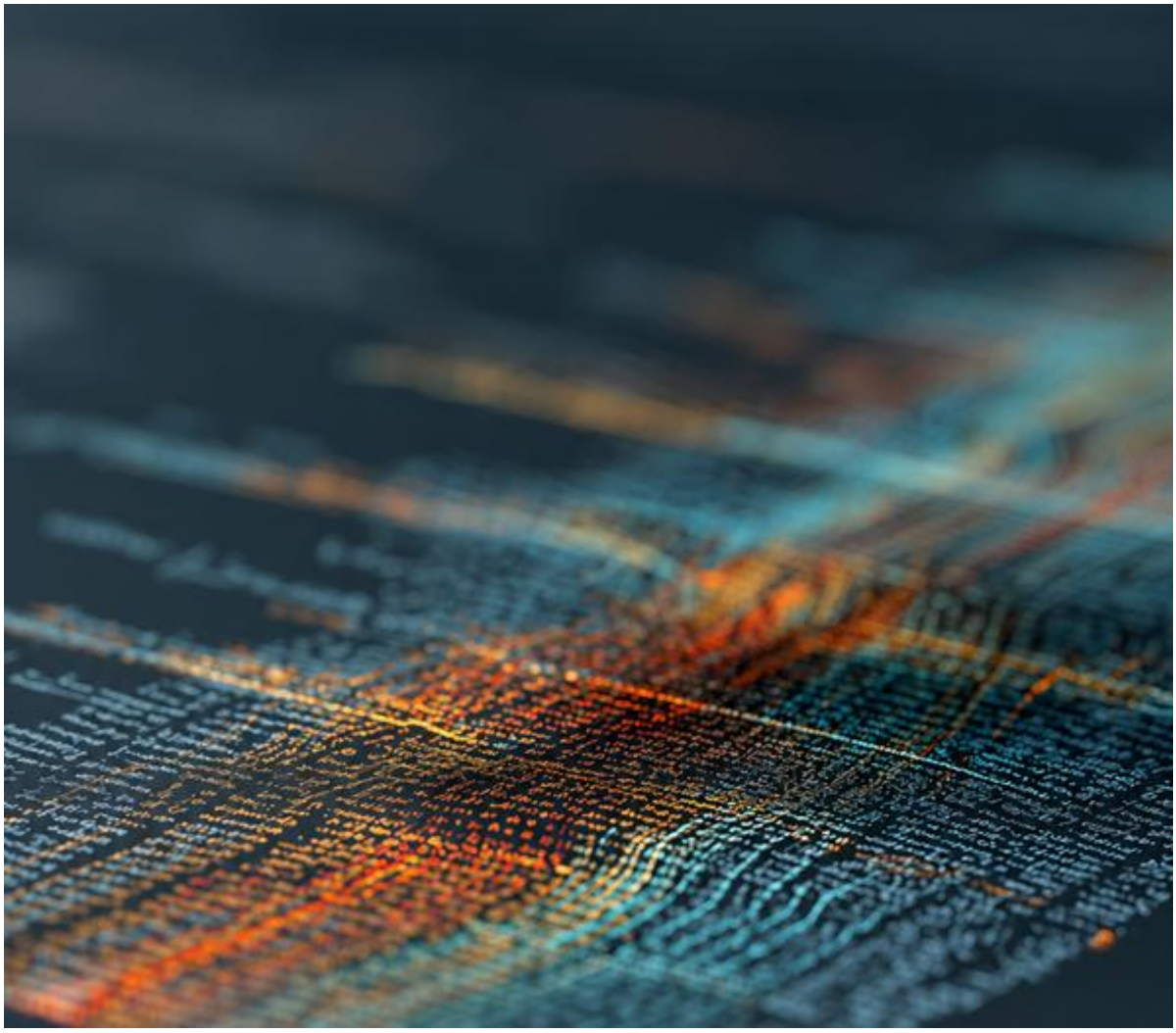
3.6.3. Risks and mitigation measures

Platform interoperability under Article 8(4) may raise semantic, technical, organisational, legal and operational risks if the scope is unclear, if the chosen model is not proportionate, or if maintenance responsibilities are weak.

Typical risks include:

- confusion between individual publication under Article 4(3) and platform-level interoperability under Article 8(4);
- selecting an overly complex model that is difficult to sustain, or an overly weak model that does not provide real interoperability in practice;
- unstable identifiers, broken links, incomplete descriptions, or inconsistent version handling;
- exposure of personal data, confidential information, or sensitive technical details through public metadata, feeds or APIs;
- unclear or inconsistent licence handling, practical exclusion of EUPL-licensed entries;
- duplicate, stale or conflicting records caused by poor synchronisation or weak update governance;
- insufficient coordination between the platform operator and the Interoperable Europe Portal team.

These risks can be mitigated by making an explicit scope and proportionality decision at the outset, defining a curated and maintainable content subset where needed, assigning stable identifiers and landing pages, carrying out an Article 8(3) conformance check before exposure, documenting field mappings and update processes, ensuring that EUPL is not blocked in practice, testing the connection before go-live, and maintaining clear operational ownership over time.



CHAPTER 4. INTEGRATING INTEROPERABILITY SOLUTIONS INTO PUBLIC PROCUREMENT

This chapter provides practical guidance on how public organisations can integrate interoperability solutions into public procurement and tenders, supporting the sharing and reuse objectives of the Interoperable Europe Act.

This chapter complements the pathways set out in Chapter 3 of these Guidelines by illustrating how procurement choices can support their implementation. It reflects the Interoperable Europe Board's role in identifying best practices for integrating interoperability solutions into public procurement and tenders under Article 15(5)(n), and the role of the Interoperable Europe Portal in listing best practices and knowledge-sharing material supporting interoperability, including guidance on public procurement, under Article 8(1)(f). This chapter focuses on interoperability-related principles derived from the IEA. It is not intended as a comprehensive guide to procurement procedures, national implementation rules or country-specific legal questions. Its purpose is to help administrations identify where procurement choices may affect later sharing, publication, adaptation or reuse.

Public procurement is one of the mechanisms through which interoperability solutions can be leveraged. Decisions taken at procurement stage influence not only the implementation of a specific interoperability solution, but also its reusability, adaptability, and long-term sustainability. By **taking interoperability considerations into account early in the procurement process**, contracting authorities can reduce duplication of efforts, avoid vendor lock-in, and facilitate future sharing and reuse.

4.1. Purpose and scope

The Interoperable Europe Act establishes two elements that are particularly relevant for procurement:

Interoperability assessments (Article 3)
Union entities and public sector bodies must carry out an interoperability assessment before taking decisions on new or substantially modified **binding requirements** that affect the cross-border interoperability of trans-European digital public services.

Sharing and reuse of interoperability solutions (Article 4)
Union entities and public sector bodies must make available interoperability solutions supporting trans-European digital public services, including relevant technical documentation and source code where applicable. When deciding on the implementation of interoperability solutions, they shall **prioritise** those **without restrictive licensing terms** where functionally equivalent.

The Regulation does not create a general obligation to conduct an interoperability assessment before every ICT procurement. However, where procurement establishes binding requirements affecting cross-border interoperability of trans-European digital public services, Article 3 may apply. Moreover, reuse and openness principles of Article 4 should inform procurement strategy and contract design.

Interoperability is often discussed in technical terms: standards, APIs, semantic models, and data formats. In practice, however, interoperability is determined by contractual choices. Every time a public authority procures:

- a case management system,
- a registry platform,
- an eHealth application,
- a digital identity interface,
- a cross-border data exchange gateway,

it makes decisions that will structure how data is stored, exchanged, licensed, modified and reused for many years.

Once contracts are signed and systems are implemented, interoperability constraints can become costly and difficult to reverse. Procurement is therefore the decisive moment at which interoperability can either be embedded, supported, or inadvertently undermined.

The IEA does not regulate procurement procedures directly. However, procurement choices may affect:

- whether an interoperability assessment is required or useful under Article 3;
- whether a procured solution can later be shared, published, adapted or maintained in line with Chapter 3 of these Guidelines;
- whether the prioritisation rule in Article 4(6) can be applied effectively where equivalent solutions are available;
- whether later reuse is facilitated or hindered by contractual rights, documentation, licensing terms.

This chapter therefore provides cross-cutting procurement guidance on planning, specifications, award criteria, contractual rights, licensing, proportionality and capacity-building.

4.2. Procurement and Interoperability Assessments

4.2.1. Rationale

Article 3 of the Act is triggered when new or substantially modified **binding requirements** affecting cross-border interoperability of trans-European digital public services are adopted. In practice, binding requirements take effect through procurement.

In the context of the Act, binding requirements may arise not only in legislation or administrative provisions, but also in contracts, calls for tender or other official documents. Procurement may therefore be relevant to Article 3 where the procurement documents set or implement such binding requirements.

At the same time, not every procurement procedure will trigger a mandatory interoperability assessment. In particular, simple procurement of standard information and communication technologies equipment, evolutive maintenance without substantive change, or routine security or technical updates will not usually require a mandatory interoperability assessment.

Where Article 3 does not require an interoperability assessment, public organisations may still carry one out voluntarily where this would support better design, planning, reuse or risk management.

4.2.2. Recommended practice

It is recommended for the contracting authorities to embed interoperability considerations directly into the early stages of procurement planning and system design, rather than treating them as a standalone compliance exercise carried out at a later stage. This can be supported by introducing an initial assessment step when defining needs and preparing technical specifications:

- it is recommended to start the assessment during needs definition and preparation of technical specifications;
- an initial screening may be carried out to determine whether cross-border data exchange or dependencies are involved;
- early identification of constraints that could affect interoperability within the Union is advisable;
- a preliminary market consultation pursuant to Article 169(1) of the Financial Regulation[8] may be helpful to gain an understanding of available market practices.

[8] Regulation (EU, Euratom) 2024/2509 on the financial rules applicable to the general budget of the Union.

Where it appears that an interoperability assessment would be beneficial, authorities are advised to define the scope of the system in a practical and proportionate manner, focusing on how the system interacts with its environment rather than producing overly complex documentation:

- it is recommended to describe interactions with other systems, stakeholders, processes, and data flows;
- authorities may map how data enters, is processed, and exits the system;
- it is advisable to determine whether the solution operates standalone or within a broader public service ecosystem.

To ensure consistency with EU-wide practices, authorities are encouraged to structure the assessment in line with the **European Interoperability Framework**, which provides a common model for analysing interoperability across multiple dimensions:

- legal: verify compliance with data-sharing rules and identify potential barriers;
- organisational: assess alignment of processes, roles, responsibilities, and governance structures;
- semantic: ensure consistent meaning of exchanged data, including use of common models or vocabularies where available, or documented mappings where they are not;
- technical: favour open standards, interoperable interfaces, and widely accepted architectures, specifications, and protocols.

To support alignment and avoid fragmentation across the Union, contracting authorities are encouraged to consider the reuse of existing European interoperability solutions or standards, in particular Interoperable Europe Solutions, before developing new ones:

- check for available EU standards, tools, and building blocks before defining bespoke specifications;
- avoid unnecessary duplication of existing solutions;
- where reuse is not pursued, it is advisable to provide a justification.

Authorities are also encouraged to consider the potential impact of the procurement on cross-border interoperability, particularly where systems may interact with services in other Member States:

- identify risks such as vendor lock-in or reliance on proprietary technologies;
- assess whether proposed solutions may hinder or enable future cross-border integration;
- define mitigation measures where relevant.

For transparency and reusability, authorities are encouraged to document the outcomes of the assessment under Article 3 in a structured but proportionate manner:

- summarise the scope and key findings across interoperability layers;
- include reuse considerations and identified risks, along with mitigation measures where appropriate;
- ensure the level of detail remains proportionate to the size and complexity of the procurement.

To ensure practical impact, authorities are encouraged to reflect interoperability considerations directly in procurement documentation, so that they become enforceable rather than purely descriptive:

- translate assessment findings into technical specifications and requirements;
- where appropriate, require the use of standards, APIs, or data portability measures.

Finally, authorities are encouraged to apply a proportionate and risk-based approach, ensuring that interoperability is considered ‘by design’ without creating unnecessary administrative burden:

- apply simplified assessments for low-risk or limited-scope procurements;

- use more detailed analysis where systems are complex or have cross-border relevance;
- ensure that interoperability considerations remain integrated into overall procurement and system design processes.

Example

A national tax authority plans to procure a new VAT reporting interface aligned with EU-level digital reporting standards. If the authority's intention is to impose mandatory technical specifications affecting cross-border interoperability, the authority should ensure an interoperability assessment has been conducted before those specifications are finalised.

4.3. Interoperability-oriented procurement design

4.3.1. Technical specifications

Technical specifications play a decisive role in determining whether a system can interoperate with others over time.

Poorly designed specifications may:

- embed proprietary data models;
- require vendor-specific integration tools;
- prevent structured data export;
- create technical isolation.

By contrast, (open) standards-based and interface-oriented specifications allow systems to:

- connect to other national systems;
- adapt to evolving EU frameworks;
- enable reuse by other public organisations.

Under the applicable procurement framework, technical specifications are to afford equal access to the procurement procedure and cannot create unjustified obstacles to competition. Where technical specifications refer to standards or equivalent references, the applicable procurement rules on equivalence continue to apply.

4.3.2. Recommended practice

Contracting authorities should:

- Refer to recognised European or international standards where available;
- Use functional specifications (e.g. 'must support structured XML/JSON data exchange') instead of brand-based descriptions;
- Require documented APIs and integration protocols;
- Ensure semantic consistency with EU data models where such models exist and are suitable, otherwise document the semantic mapping, correspondence table or justification used.

The objective is not to prescribe a particular technology but to ensure systems can communicate, evolve and integrate with other systems.

Example

When procuring a data platform, the contracting authority could require compliance with existing EU eProcurement data standards, ensuring that future cross-border analytics and reporting are feasible.

4.4. Preventing vendor lock-in

4.4.1. Rationale

Vendor lock-in occurs when switching suppliers, maintainers, or delivery models becomes technically, legally, or economically difficult. It is considered harmful, because it restricts digital competition, stifles innovation, and threatens economic sovereignty by making businesses and public entities dependent on specific providers. The risk may be particularly acute where central or business-critical infrastructures are developed or operated under commercial arrangements that do not secure sufficient rights, documentation, portability, governance transparency or exit readiness for the public authority concerned.

Lock-in is thus also incompatible with the objectives of interoperability because it:

- restricts reuse;
- increases long-term costs;
- limits competition;
- undermines data portability.

It is important to note that interoperability in this context is not only about cross-border compatibility but also about preserving public authorities' technological autonomy. It also concerns operational autonomy, knowledge retention, data portability and the practical ability to transition to another maintainer or provider without disproportionate disruption.

4.4.2. Recommended measures

Contracts should include:

- data export obligations in structured, documented formats;
- documentation of all interfaces, customisations, and key configuration choices;
- clear licensing and intellectual property terms;
- knowledge-transfer obligations covering architecture, configuration, build, deployment and operating procedures, where relevant
- exit and transition assistance clauses;
- rights and practical arrangements enabling a replacement maintainer or provider to operate, maintain, adapt and migrate the solution;
- provisions requiring compatibility maintenance during updates.

Example

In a contract for a case management system, the authority may require:

- annual delivery of a full data export;
- documentation of all schema changes;
- support for migration during contract termination.

These clauses reduce the risk that interoperability is lost at contract end.

4.5. Award criteria

4.5.1. Rationale

Minimum interoperability requirements set out in technical specifications can help establish baseline compliance, but do not guarantee long term adaptability, cross-border scalability, ease of reuse by other public authorities, avoidance of lock-in or sustainability over time. Award criteria allow authorities to differentiate between basic compliance and solutions that actively enhance interoperability in a robust, future-proof and reusable manner.

If interoperability is not reflected in award criteria, suppliers have limited incentive to go beyond basic conformity. By contrast, when interoperability is rewarded, the market is encouraged to internalise public policy objectives.

4.5.2. General principles for designing interoperability-related award criteria

Interoperability-related award criteria should:

- be **linked to the subject matter** of the contract, and demonstrably contribute to the quality, performance or functional characteristics of the procured solution;
- be **objectively measurable**, with clearly defined evaluation parameters enabling consistent assessment across bidder;
- respect the principles of **transparency, equal treatment** and **non-discrimination**;
- allow for **equivalent solutions**, including different technical approaches capable of achieving comparable interoperability outcomes;
- be **proportionate** to the contract's nature, scope and strategic importance.

Award criteria should be formulated in a way that enables verifiable and reproducible evaluation. Vague or discretionary formulations (e.g. 'innovative interoperability approach') without defined assessment parameters should be avoided. Instead, award criteria should identify the specific aspects to be assessed and, where appropriate, the expected level of performance.

To ensure transparency and legal certainty, contracting authorities should specify in the procurement documents:

- the **evaluation elements** that will be assessed under each criterion;
- the **evidence to be provided** by bidders (e.g. technical documentation, references, access to test environments);
- the **scoring methodology**, including how qualitative judgments will be translated into scores;
- the **weighting** of interoperability-related criteria within the overall award framework, ensuring that it reflects their relevance to the contract.

Contracting authorities may combine mandatory minimum interoperability requirements with scored quality criteria. Where a feature is indispensable for lawful, secure or workable integration, for example the existence of usable API documentation, it may be preferable to set a pass/fail baseline in the technical specifications or procurement documents and reserve comparative scoring for quality features that go beyond that baseline.

In order to ensure consistency and transparency, interoperability-related award criteria should be accompanied by predefined scoring scales with clearly described performance levels. A commonly used approach is a **0–5** or 0-10 **scoring scale**, which may be defined as follows:

- **0 – Non-compliant / not demonstrated**: The proposal fails to address the criterion or provides insufficient or irrelevant information;

- **1 – Very low quality**: The proposal addresses the criterion only minimally, with significant gaps or unclear elements;
- **2 – Basic quality**: The proposal meets minimum expectations but lacks completeness, robustness or supporting evidence;
- **3 – Satisfactory quality**: The proposal adequately meets the criterion, with reasonable detail and no major deficiencies;
- **4 – Good quality**: The proposal demonstrates a high level of quality, with well-developed, credible and substantiated elements;
- **5 – High quality**: The proposal demonstrates a comprehensive, well-documented and mature approach that significantly exceeds baseline expectations and provides clear added value in terms of interoperability.

However, scoring scales should not be applied mechanically across different criteria. Contracting authorities are encouraged to define criterion-specific descriptors that reflect the subject matter of the contract.

Example

Applying the 0–5 scoring scale to **API documentation**:

- *Score 0*: No API documentation provided, or documentation is inaccessible or irrelevant;
- *Score 1*: The documentation is minimal and fragmented, providing only very limited information such as an incomplete list of endpoints. Key elements such as data structures, authentication methods, or error handling are missing or insufficiently described, making the API difficult to understand or use;
- *Score 2*: The documentation covers the main endpoints and provides some guidance on usage, but remains incomplete. Descriptions of request and response formats, error handling, or authentication are only partially included or lack clarity, which may limit effective integration;
- *Score 3*: The documentation is sufficiently complete and structured, covering core endpoints, data models, authentication mechanisms, and basic error handling in a clear and usable way. However, it may lack advanced features such as extensive examples, detailed explanations, or enhanced usability tools;
- *Score 4*: The documentation is comprehensive and well-structured, including full endpoint coverage, clearly defined data models, detailed error handling, and practical usage examples. It also includes information on versioning and may provide access to a sandbox or test environment, supporting effective integration and testing;
- *Score 5*: The documentation is fully comprehensive and developer-oriented, following recognised standard. It includes clear versioning and backward compatibility policies, interactive or executable documentation, access to a fully functional sandbox, and strong usability features such as search, tutorials, and examples, supported by evidence of ongoing maintenance and updates.

Example

Applying the 0–5 scoring scale to **data portability**:

- *Score 0:* No data export functionality is available, or data cannot be extracted without disproportionate effort, effectively preventing portability;
- *Score 1:* Data export is extremely limited and largely manual, requiring significant effort and lacking defined structure, standard formats, or supporting documentation, which makes data extraction difficult and inefficient;
- *Score 2:* Data export is available in some structured formats, but the scope is limited. Metadata is incomplete, processes may be partly manual, and documentation is insufficient to ensure straightforward or reliable migration;
- *Score 3:* Data can be exported in structured, machine-readable formats, with a reasonable level of completeness and associated metadata. Procedures for export are documented, although automation and supporting tools may be limited;
- *Score 4:* Data export is automated or easily executable and provided in widely used, standard formats. It is accompanied by comprehensive metadata, clear documentation, and, where applicable, dedicated migration tools. The process is efficient and does not require disproportionate effort;
- *Score 5:* Data portability is fully supported through automated, end-to-end export mechanisms using standardised and interoperable formats. Complete metadata is provided, along with dedicated migration toolkits or APIs, thoroughly documented and tested migration procedures, and demonstrably low-cost, low-effort transition processes, potentially supported by evidence of successful prior migrations.

Where appropriate, scoring may also be incorporated:

- **thresholds or minimum acceptable scores** for critical interoperability aspects;
- **sub-criteria** to break down complex topics (e.g. documentation, versioning, testing under interface quality);
- **quantifiable indicators** (e.g. number of supported standards, availability of specific features), provided these remain proportionate and do not unduly restrict competition.

The scoring methodology should be sufficiently precise to allow evaluators to justify the scores awarded and enable bidders to understand how their proposals will be assessed.

4.5.2.1. Openness and standards implementation

Objective: Reward robust and standards-based implementation beyond minimum requirements.

Rationale: Standards reduce fragmentation and increase cross-border interoperability. Awarding additional points encourages suppliers to design beyond baseline compliance in ways that strengthen practical interoperability.

Possible evaluation elements:

- extent of compliance with recognised European or international standards;
- degree of alignment with relevant EU-level data models or, where no such model is available, the clarity and quality of the semantic mapping proposed;
- use of widely adopted open specifications;
- avoidance of proprietary protocols where alternatives exist.

4.5.2.2. Quality and interface maturity

Objective: Promote technically mature and well-documented system interfaces.

Rationale: Many interoperability failures stem not from absence of interfaces but from poor documentation and weak lifecycle management.

Evaluation elements may include:

- availability of fully documented APIs;
- clarity and completeness of interface documentation;
- versioning policy;
- backward compatibility management;
- availability of sandbox/test environments;
- conformance testing mechanisms.

Example

In a procurement for a cross-border registry interface, bidders could be scored on the clarity of API documentation and the robustness of version management policies.

4.5.2.3. Data portability and migration tools

Objective: Strengthen long-term sustainability and prevent vendor lock-in.

Rationale: Interoperability includes the ability to exit and transition systems without prohibitive cost.

Evaluation criteria may consider:

- automated data export capabilities;
- use of structured, machine-readable formats;
- completeness of metadata;
- availability of migration toolkits;
- estimated cost and duration of data extraction at contract end.

4.5.2.4. Licensing conditions and reuse facilitation

Objective: Ensure that licensing and intellectual property arrangements enable reuse, adaptation, and sharing of solutions across public organisations.

Rationale: Reuse cannot occur if legal rights are absent or overly restrictive.

Award criteria may assess:

- clarity and transparency of intellectual property arrangements;
- extent to which the proposed licensing model permits reuse by other public sector bodies;
- absence of restrictive reuse conditions;
- ability to modify and adapt deliverables;
- extent to which the proposed delivery model supports operational autonomy, knowledge retention, exit readiness and transition to alternative maintainers or providers.

In a procurement for a cross-border registry interface, bidders could be scored on the clarity of API documentation and the robustness of version management policies.

This does not mandate a specific licensing model. However, when two technically equivalent solutions are proposed, authorities may award additional points to solutions that:

- are supported by more than one single provider;
- allow broader reuse rights;
- minimise legal barriers to adaptation;
- facilitate handover to alternative maintainers or providers;
- support publication, dissemination, and lawful reuse.

4.5.2.5. Modularity and architectural flexibility

Objective: Ensure systems can evolve and integrate with future EU frameworks.

Rationale: Modularity supports adaptability and reduces long-term fragmentation.

Evaluation elements:

- modular system design;
- clear separation of components;
- use of interoperable microservices;
- replaceability of components without systemic disruption.

Example

In a digital identity infrastructure procurement, additional points may be awarded to solutions allowing independent replacement of authentication modules without reengineering the entire system.

4.5.2.6. Demonstrated interoperability experience

Objective: Recognise and reward proven experience in delivering interoperable solutions, particularly in cross-border or multi public organisations contexts.

Rationale: Demonstrated experience in implementing interoperable systems reduces implementation risk and increases confidence in the supplier's ability to deliver complex solutions.

Award criteria may consider:

- proven deployment in other Member States;
- evidence of cross-border data exchange in comparable contexts;
- participation in EU interoperability initiatives;
- prior reuse of the solution by multiple public organisations.

It is recommended for this criterion to be applied carefully to avoid discrimination and should focus on demonstrated technical capacity rather than geographic origin.

4.5.2.7. Governance and interoperability management

Objective: Ensure that appropriate governance mechanisms are in place to manage interoperability throughout the lifecycle of the system.

Rationale: Strong governance ensures that interoperability is maintained over time and adapts to evolving requirements and standards.

This is relevant both for solutions developed in-house and for solutions delivered through commercial arrangements. In both cases, long-term reuse depends on clear ownership, maintainership, documentation continuity, and the practical ability to avoid dependency on a single supplier or operator.

In practice, the long-term viability of a reusable solution often depends not only on the technical artefacts themselves, but also on a credible maintenance and governance model, a clearly identified maintainer or coordinating function, and, where relevant, an active ecosystem of users or contributors capable of supporting its continued improvement over time. This applies equally to open source and non-open source solutions.

Criteria may assess:

- sustainability of maintenance and release governance;
- proposed change management procedures;
- clarity of maintainer responsibilities, escalation paths and stewardship arrangements;
- existence of a credible user, maintainer or contributor ecosystem, where relevant to the solution type;
- standards update monitoring processes;
- interoperability compliance testing strategies;
- long-term maintenance of interfaces;
- arrangements for knowledge retention, documentation continuity and handover.

Example

Additional points may be awarded where bidders propose structured monitoring of evolving EU interoperability and a commitment to implement updates without additional licensing costs.

4.6. Sharing obligation and intellectual property rights

4.6.1. Rationale

> **Article 4 (1)**
>
> Interoperability solutions supporting trans-European digital public services are to be made available to other public sector bodies upon request.
>
> **Article 4 (1) (b)**
>
> The obligation to share shall not apply to interoperability solutions for which third parties hold intellectual property rights that restrict the possibility of sharing the solution for reuse.

Union entities and public sector bodies are required, under the conditions laid down in Article 4(1), to make interoperability solutions supporting trans-European digital public services available to another Union entity or public sector body that requests them. However, this obligation does not apply where making available an interoperability solution would infringe intellectual property rights or licensing conditions agreed with third parties.

Contracts should therefore be drafted to secure rights that enable sharing and reuse, while also identifying and appropriately managing cases where exceptions apply.

4.6.2. Recommended contractual clauses

Contracts should:

- clarify ownership of deliverables;

- secure rights to share and reuse solutions, including adaptation by other public organisations where applicable;
- require delivery of technical documentation, including manuals and documented code;
- ensure source code access where appropriate;
- define clearly whether deployment assistance, onboarding, hosting, support, trial access or other operational services are included, optional, or excluded;
- where relevant, distinguish the rights in the interoperability solution itself from any separate paid service or support components.

4.7. Prioritising solutions without restrictive licensing

Example

A Member State develops a cross-border licensing interface. The contract should explicitly allow:

- publication of the solution on the Interoperable Europe portal;
- adaptation by another Member State;
- sharing of updated versions.

Without such clauses, reuse may be legally impossible.

4.7.1. Rationale

The Act requires Union entities and public sector bodies to prioritise interoperability solutions without restrictive licensing terms where those solutions are equivalent in the sense of Article 4(6).

The reason for that provision is structural:

- restrictive licences may prohibit sharing;
- they may limit adaptation;
- they may prevent cross-border reuse.

Where two solutions offer equivalent functionality, the one allowing reuse without restrictive conditions better supports interoperability and should be prioritised.

4.7.2. Practical application

Procurement evaluation may include:

- analysis of licensing terms;
- assessment of reuse rights;
- consideration of long-term access conditions.

It is important to note that prioritizing solutions without restrictive licensing should not be understood as mandating an open source procurement. It requires prioritisation where equivalent solutions without restrictive licensing terms are available, but it does not prescribe a specific procurement technique for giving effect to that rule.

4.7.3. Equal treatment and non-discrimination

Interoperability requirements should not be designed in a manner that:

- favours specific vendors or technologies without objective justification;
- indirectly exclude certain economic operators without proportionate reasons.

Where open standards are referenced, authorities are to allow equivalent solutions and provide objective criteria for assessing equivalence. This is of particular importance in the context of open

source solutions, whose promotion within the framework of a procurement procedure cannot be conducted to the general exclusion of non-open source solutions.

4.8. Proportionality

> Art 18(1) of the Directive 2014/24/EU states that "contracting authorities shall treat economic operators equally and without discrimination and shall act in a transparent and **proportionate** manner".

Interoperability requirements should also be proportionate to the subject matter and value of the contract.

Authorities should ensure that:

- requirements do not impose unnecessary administrative or technical burdens;
- certification or documentation requirements are limited to what is necessary;
- small and medium-sized enterprises are not unduly excluded.

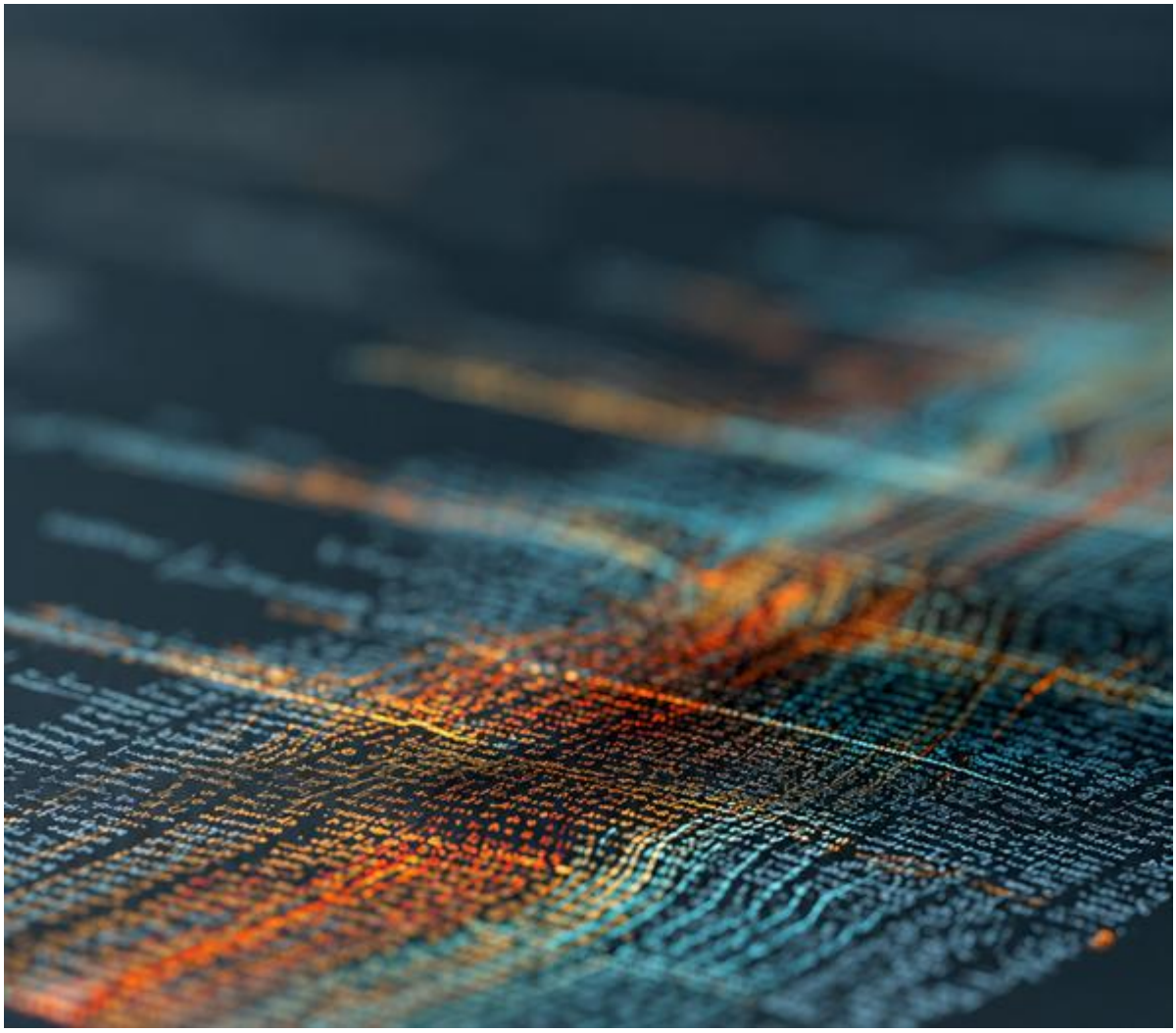
4.9. Capacity building and monitoring

It is advisable that Union entities and public sector bodies:

- provide training for procurement officials on interoperability;
- develop model clauses and evaluation templates;
- establish advisory mechanisms for complex ICT procurements;
- monitor the extent to which procurement procedures incorporate interoperability requirements.

Monitoring may include indicators such as:

- the proportion of ICT procurements including interoperability criteria;
- the degree of reuse of existing solutions;
- the reduction of proprietary dependencies.



ANNEXES

Annex 1: Further Resources

Annex 2: Principles for Sharing Interoperability Solutions

Annex 3: Maintenance Responsibilities under the Interoperable Europe Act

Annex 4: Metadata and Publication Context (Operational Overview)

Annex 5: Voluntary Best Practices for Publishing Interoperability Solutions

Annex 6: Indicative Structure for Documenting an Adapted Interoperability Solution

ANNEX 1: FURTHER RESOURCES

Legal and regulatory framework

Binding legal and regulatory context for these Guidelines:

- Regulation (EU) 2024/903 of the European Parliament and of the Council of 13 March 2024 laying down measures for a high level of public sector interoperability across the Union (Interoperable Europe Act) [URL].
- Directive 2014/24/EU of the European Parliament and of the Council of 26 February 2014 on public procurement and repealing Directive 2004/18/EC Text with EEA relevance [URL].

Core interoperability policy frameworks

EU policy instruments defining interoperability concepts, principles and analytical models.

- European Interoperability Framework (European Commission, 2017) [URL].
- Guidelines for Interoperability Assessments (European Commission, 2024) [URL].

Sharing, reuse and open source policy guidance

Guidance on sharing, reuse and open source practices in the public sector.

- OSOR Handbook – Open Source Software in Public Administration (European Commission: DG DIGIT, 2026) [URL].
- Guidelines for Sustainable Open Source Communities in the Public Sector (European Commission: DG DIGIT, 2021) [URL].
- Open Source Software Adoption and Reuse in European Local Governments (European Commission: DG DIGIT, 2025) [URL].
- The Sharing and Reuse Framework for IT Solutions (European Commission, 2017) [URL].
- European Union Public Licence – Welcome (Interoperable Europe Portal, 2025) [URL]

Operational tools and support instruments under the European Interoperability Framework

Practical tools and institutional mechanisms supporting implementation of the Act and the Guidelines.

- Common Assessment Method for Standards and Specifications (CAMSS) (Interoperable Europe Portal, 2026) [URL]
- Interoperable Europe Academy Courses (Interoperable Europe Portal) [URL]
- Licensing Assistant: Compatibility Checker (Interoperable Europe Portal, 2026) [URL]
- EC Open Source Programme Office (EC OSPO) (Interoperable Europe Portal) [URL]

Interoperable Europe Portal, catalogues and knowledge repositories

Resources supporting discovery, publication, visibility and governance of interoperability solutions.

- Interoperable Europe Portal (2026) [URL]
- EU Interoperability Solutions Catalogue (Interoperable Europe Portal, 2026) [URL]
- Eligibility criteria (Interoperable Europe Portal, 2026) [URL]
- How to create Solutions on the Portal (Interoperable Europe Portal, 2024) [URL]
- Interoperable Europe Solutions (Interoperable Europe Portal, 2026) [URL]

- EU Open Source Solutions Catalogue (Interoperable Europe Portal, 2026) [URL]
- Interoperable Europe Board Documents: Meeting Minutes (Interoperable Europe Portal, 2024 onwards) [URL]
- Digital Public Administration Factsheets and Supporting Documents (European Commission: DG DIGIT, 2024) [URL]

Practical case studies and implementation examples

- Annex A: Guide to publishing software as open source (Docs Italia) [URL]
- Collaborative software solutions together with Docs (OpenDesk, 2025) [URL]
- In-house developments' Munich Open Source [URL]
- Everything you need for meaningful citizen participation (Consul Democracy) [URL]
- 'Open Source Observatory: Case Studies', Interoperable Europe Portal, 2019 onwards [URL]
- OS2's governance model (OS2) [URL]
- Oskari – a mapping tool that adapts to your needs (Oskari) [URL]
- Who we are (iMio) [URL]
- X-Road implementation models (X-Road Global) [URL]
- Open Source Observatory: News and Events (Interoperable Europe Portal, 2025 onwards) [URL]

ANNEX 2: PRINCIPLES FOR SHARING INTEROPERABILITY SOLUTIONS

The sharing of interoperability solutions is guided by the principles derived from the Interoperable Europe Act, which aims to enable the cross-border interoperability of trans-European digital public services and contribute to efficiency, cooperation and the avoidance of duplication. These principles underpin both the development of interoperability solutions and the conditions under which such solutions are shared and reused, with a view to achieving a high level of public sector interoperability across the European Union.

- **Cooperation and reuse**

  Public organisations should cooperate in the development and sharing of interoperability solutions and prioritise reuse over isolated development. Sharing and reuse support cross-border and cross-sectoral cooperation, reduce fragmentation, and contribute to the efficient use of public resources.

- **Openness and availability**

  Interoperability solutions should be made available in a manner that facilitates access and reuse by other public organisations, wherever legally and technically possible. Openness supports transparency and cooperation, subject to applicable limitations and safeguards laid down in the IEA.

- **Security, confidentiality, and data protection**

  The sharing of interoperability solutions should comply with applicable requirements on security, confidentiality and data protection. Public organisations should ensure that shared solutions and accompanying documentation do not disclose sensitive information and that appropriate safeguards are in place to preserve integrity, availability, authenticity, and trust in cross-border digital public services.

- **Transparency and legal clarity**

  Public sector bodies should ensure transparency regarding the conditions for sharing and reuse of interoperability solutions. Clear information on solution functionality, licensing, dependencies and governance should be made available to support informed reuse and to ensure legal certainty and respect for intellectual property rights.

- **Sustainability and efficient use of resources**

  Interoperability solutions should be shared and reused in a manner that supports their sustainability over time, including consideration of maintenance and further development. Where appropriate, cooperative or cost-sharing arrangements may support the efficient and sustainable use of public resources.

- **Interoperability by design**

  Interoperability considerations should be embedded from the earliest stages of design and development of digital public services and solutions. Interoperability solutions should be designed to facilitate cooperation, reuse and integration across organisational and territorial boundaries, without creating unnecessary barriers to cross-border interoperability.

ANNEX 3: MAINTENANCE RESPONSIBILITIES UNDER THE INTEROPERABLE EUROPE ACT

This Annex clarifies how maintenance responsibilities are allocated under different sharing pathways.

Direct sharing of interoperability solution upon request (Article 4(1) and 4(2)) (Section 3.3)

Legal position

When an interoperability solution is shared directly upon request under Article 4(1):

- The sharing entity must make the solution available, including technical documentation and, where applicable, version history, documented source code and references to open standards or technical specifications used.
- Under Article 4(2), the sharing entity may specify conditions that apply to reuse, including any guarantees regarding cooperation, support and maintenance.
- The Act does not require the sharing entity to provide ongoing maintenance, support, or upgrades.

Practical consequence

- Maintenance is provided only if the sharing entity voluntarily offers it and specifies it in the reuse conditions.
- If no maintenance or support is offered, this should be stated clearly.
- Once adopted, the reusing entity will normally need to organise operation, maintenance and evolution of the solution in its own environment, subject to the reuse conditions.

Publishing interoperability solutions via the Interoperable Europe Portal or connected portals, catalogues, or repositories (Article 4(3) and Article 8(3)) (Section 3.2)

Legal position

Where the sharing obligation is fulfilled by publication under Article 4(3), and the solution is accessible through the Interoperable Europe Portal:

- Article 4(2) does not apply.
- Under Article 8(3)(e), solutions accessible through the Portal must be regularly maintained under the responsibility of the owner of the interoperability solution.

Practical consequence

- The owner of a Portal-accessible solution is responsible for maintenance of the solution within the meaning of Article 8(3)(e).
- Maintenance responsibility includes keeping the published information accurate (i.e. keeping the solution's reference information, links and lifecycle status accurate, and indicating clearly when the solution is superseded, inactive or no longer maintained).
- If maintenance is discontinued, the owner should update the maintenance status and assess whether the solution should remain accessible through the Portal.

The Act does not define a specific maintenance frequency or service level. Maintenance should remain proportionate to the nature of the solution and consistent with Article 8(3).

Sharing adapted interoperability solutions (Article 4(4)) (Section 3.4)

Legal position

Under Article 4(4):

- A reusing entity may adapt an interoperability solution, unless restricted by third-party intellectual property rights.
- If the original solution was made public under Article 4(3), the adapted solution must be made public in the same way.

Practical consequence

- The original owner is not responsible for maintaining adaptations created by other entities.
- Where the adapted solution is made accessible through the Portal, the maintenance responsibility under Article 8(3)(e) applies to the owner of that adapted solution.

Cost-sharing agreements for future developments of interoperability solutions (Article 4(5)) (Section 3.5)

Legal position

Article 4(5) allows sharing and reusing entities to conclude an agreement on sharing the costs for future developments of an interoperability solution.

The Act does not prescribe how maintenance must be allocated in such agreements.

Practical consequence

- Maintenance responsibilities in cost-sharing arrangements are defined contractually.
- Agreements may allocate maintenance tasks through financial contributions, in-kind contributions (for example human resources), or shared governance structures.
- Where an obligation to share exists under Article 4(1), cost-sharing arrangements for future developments do not affect the sharing of the existing interoperability solution.

ANNEX 4: METADATA AND PUBLICATION CONTEXT (OPERATIONAL OVERVIEW)

This Annex explains the role of metadata in the context of the Interoperable Europe Act and its use on the Interoperable Europe Portal.

The IEA does not impose a single metadata standard for publishing or sharing interoperability solutions. However, different publication and interoperability contexts rely on different types of descriptive information in practice. Understanding these distinctions helps public organisations apply proportionate measures and avoid unnecessary complexity.

Where a solution is made accessible through the Interoperable Europe Portal, the conditions set out in Article 8(3) of the Act apply independently of any metadata format used.

Publishing directly on the Interoperable Europe Portal

(Article 4(3))

When a sharing entity publishes a solution directly on the Interoperable Europe Portal, it needs to provide the information required by the current Portal submission workflow[9] in order to be published:

- These fields are defined as part of the Interoperable Europe Portal's operation.
- They support search, discovery and understanding by users.
- They do not constitute a legally mandated metadata standard under the IEA.

In this case:

- no specific metadata schema (such as ADMS-AP or publiccode.yml) is legally required for direct publication;
- a solution published on the Portal may be hosted directly within the Portal's solution space and may also refer to external repositories or websites, depending on how the owner organises the solution.

Publishing via a Connected portal, registry, catalogue, or version-controlled repository

(Article 4(3), connected publication route)

Where a solution is published via a portal, catalogue or repository connected to the Interoperable Europe Portal, the connected route should ensure that the exposed entry can be referred to or made accessible through the Portal in a stable and maintainable manner. In practice, this may rely on stable landing pages, structured harvested records, or a combination of both, depending on the connection model.

In practice:

- basic descriptive information should be available so that users can understand and reuse the solution;
- the connection may be implemented through simple linking, manual curation, or more automated mechanisms.

[9] How to create Solutions on the Portal | Interoperable Europe Portal,
How-To Articles | Interoperable Europe Portal,
Eligibility criteria | Interoperable Europe Portal

The IEA does not mandate machine-readable metadata in this case. However, structured metadata becomes relevant where automated synchronisation or federation is desired.

Connecting portals, catalogues and repositories with the Interoperable Europe Portal

(Article 8(4))

This applies when a Union entity or public sector body operates a portal, catalogue or repository with similar functions and must take necessary and proportionate measures to ensure interoperability with the Interoperable Europe Portal.

In this context:

- interoperability typically requires the exchange of structured, machine-readable descriptive information;
- the Act does not prescribe a single metadata schema, but automated federation and large-scale synchronisation are typically facilitated by structured metadata and agreed mappings.

In practice:

- ADMS-AP is used to describe a wide range of interoperability solutions (not limited to software), supporting discoverability and federation across platforms;
- ADMS-AP has historically been used within the Interoperable Europe Portal ecosystem to describe interoperability assets;
- DCAT-AP may be more suitable where the connected platform is primarily a data catalogue.

Open source solutions and the EU OSS Catalogue

(operational practice)

The EU OSS Catalogue is a catalogue hosted within the Interoperable Europe Portal ecosystem and operates on a federated model.

For open source software solutions:

- the EU OSS Catalogue uses publiccode.yml as its principal metadata specification;
- onboarding and automated synchronisation also require, in practice, a public git repository and a publicly accessible API.

These requirements:

- are operational requirements of the EU OSS Catalogue;
- do not create additional legal obligations under the IEA;
- should not be generalised to all interoperability solutions or all Portal connection models;
- determine whether automated federation with the EU OSS Catalogue is possible in practice.

IN SUMMARY

- Portal publication: complete Portal workflow fields; no metadata standard required by law.
- Connected publication without federation: stable references and clear descriptive information may be sufficient.
- Open source software federation via the EU OSS Catalogue: publiccode.yml, a public git repository and a public API are currently required in practice.

ANNEX 5: VOLUNTARY BEST PRACTICES FOR PUBLISHING INTEROPERABILITY SOLUTIONS

This Annex outlines voluntary best practices that may enhance the quality, clarity, sustainability and reusability of interoperability solutions published on the Interoperable Europe Portal.

These practices go beyond the minimum requirements of Articles 4 and 8 of the IEA. They are optional and intended as inspiration for public organisations seeking to maximise the impact of their solutions.

Clarity and usability of documentation

Clear and structured documentation increases the likelihood that other public organisations will successfully reuse a solution. Public organisations may consider:

- providing a concise overview explaining purpose and scope;
- briefly describing the solution's main features or main functional capabilities;
- including basic installation or integration guidance;
- presenting architectural overviews for complex systems;
- organising documentation in a way that reflects real-world implementation steps.

The level of detail should remain proportionate to the complexity of the solution.

Interoperability and alignment

Where relevant, solutions may benefit from alignment with open and widely used standards, formats and interface conventions. Public organisations may consider:

- documenting APIs in a clear and structured manner;
- clarifying data models and semantic structures;
- referencing recognised standards or interoperability frameworks where applicable.

Such alignment is not mandatory under the Act but may reduce integration complexity for reusing entities.

Security-aware publication

Before publication, public organisations may review materials to ensure that no sensitive information is disclosed. This may include:

- removing credentials, keys, tokens or internal configuration details;
- verifying third-party dependencies and update status;
- providing a contact point for reporting security issues.

These practices support responsible reuse but do not introduce additional legal obligations.

Governance transparency

Solutions benefit from clarity about ownership and decision-making. Public organisations may indicate:

- who maintains the solution;
- how updates are managed;
- how feedback or contributions may be submitted.

Where multiple organisations are involved, transparent governance arrangements can help ensure continuity. The appropriate model will vary according to the solution: some solutions may rely on a single owner, others on a consortium, community or foundation.

Lifecycle communication

Communicating the lifecycle stage of a solution helps reusing entities make informed decisions.

Public organisations may indicate whether a solution is:

- under active development,
- stable,
- limited to security updates,
- deprecated or end-of-life.

Providing version identifiers and change summaries can further enhance transparency.

Encouraging adoption and engagement

Publication alone does not guarantee reuse. Public organisations may support adoption by:

- announcing new releases;
- sharing lessons learned;
- responding to feedback;
- participating in relevant networks.

Even modest engagement can strengthen cross-border collaboration.

Advanced operational practices (optional)

Public organisations with greater technical capacity may adopt advanced practices such as:

- automated testing and integration pipelines;
- structured release management;
- reproducible environments;
- sustainability planning for long-term maintenance.

These practices are entirely optional and reflect varying levels of organisational maturity.

ANNEX 6: INDICATIVE STRUCTURE FOR DOCUMENTING AN ADAPTED INTEROPERABILITY SOLUTION

This Annex provides an indicative structure that public organisations may use when documenting an adapted interoperability solution.

Public organisations should use this structure in a proportionate manner.

For a limited adaptation, it may be sufficient to document only:

- the source solution;
- the target context;
- the main changes made;
- any key legal, technical, or interoperability implications.

For more substantial adaptations, public organisations may find it useful to complete all sections of this Annex in a more structured way.

Identification of the adapted solution

This section should identify the adapted interoperability solution clearly and unambiguously. It may include:

- the title or name of the adapted solution;
- the type of solution, such as software, semantic artefact, specification, documentation, or a mixed package;
- the public organisation or third-party responsible for the adaptation;
- the date and version of the adapted solution;
- a stable identifier, reference, or repository link, where available.

Upstream solution and version used

This section should describe the upstream solution on which the adaptation is based. It may include:

- the name of the upstream solution;
- a stable reference, repository link, or Interoperable Europe Portal entry;
- the version, release, or tag used as the basis for the adaptation;
- the original owner, maintainer, or responsible entity;
- the applicable licence or licensing terms of the upstream solution.

Where relevant, this section may also indicate whether the upstream solution was made public under Article 4(3).

Purpose and target context of the adaptation

This section should explain why the adaptation was made and in what context it is intended to be used. It may include:

- the service, policy, administrative, or operational context of the adaptation;
- the intended users or reusers of the adapted solution;
- the main reason why the upstream solution could not be reused unchanged;
- autonomy assessment required by the sharing entity of the upstream solution;
- any relevant national, regional, sectoral, or organisational context.

Summary of the main adaptations made

This section should explain what was changed in the adapted solution. It may include:

- a short summary of the modifications made;
- the rationale for the adaptation;
- a distinction, where useful, between changes to different artefacts, such as software, data models, specifications, documentation, interfaces, or governance arrangements;
- a short indication of whether the adaptation remains closely aligned with the upstream solution or introduces more substantial divergence.

Implications for interoperability, deployment, or compliance

This section should describe any significant implications resulting from the adaptation. It may include:

- implications for interoperability with the upstream solution or with other systems;
- implications for deployment or integration;
- implications for legal or regulatory compliance, including where relevant cybersecurity, availability, or data protection;
- any relevant dependencies, limitations, or constraints that future reusers should be aware of.

Licensing, rights, and publication considerations

This section should clarify the legal framework applying to the adapted solution. It may include:

- the licence applying to the adapted solution;
- any relevant third-party intellectual property rights considerations;
- whether the adaptation may be shared or published and under what conditions;
- where relevant, whether publication of the adapted solution is required because the upstream solution was made public under Article 4(3).

Maintenance, governance, and relationship to the source solution

This section should explain how the adapted solution will be maintained and how it relates to the upstream solution over time. It may include:

- the entity or team responsible for maintenance;
- whether the adapted solution will track upstream changes, remain aligned, or diverge over time;
- how updates, bug fixes, or security patches from the upstream solution will be monitored and handled;
- whether contributions back to the upstream solution are envisaged;
- a contact point for questions on reuse, where appropriate.

Risks or dependencies, where relevant

This section may be used to record any key risks, dependencies, or issues affecting reuse. It may include:

- technical, legal, organisational, or semantic dependencies;
- risks linked to divergence from the upstream solution;
- licensing or compatibility risks;
- maintenance or sustainability risks;
- any mitigation measures, where useful.

